



The **AiEdge** ACADEMY

Training LLMs to Follow Instructions



Training an LLM to Follow Instructions

- **Pretraining (language modeling)**
- **Supervised Fine-tuning (to follow instructions)**
- **Reinforcement Learning (to understand the difference between a good and bad answer)**



Training an LLM to Follow Instructions

InstructGPT – 2022

Training language models to follow instructions with human feedback

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OpenAI



Training an LLM to Follow Instructions

Pretrained model

Prompt

Output

"How to be productive..."

Learn the structure of language and general knowledge by predicting the next word in a sequence of words.

- General advice, based on common knowledge.
- Broad and not tailored to any specific context or detailed strategies.
- The model regurgitates information it has seen during pre-training, which is generic.

"In order to be productive, it's important to stay focused on your goals. Being productive involves managing your time efficiently, prioritizing tasks, and avoiding distractions. It's also helpful to take regular breaks to recharge and maintain a healthy work-life balance."



Training an LLM to Follow Instructions

Supervised Learning Fine-tuning

Prompt

Output

"How to be productive..."

Adapt the model to generate text that is more specific or tailored to particular styles or types of tasks, based on labeled datasets.

- More specific strategies, possibly reflecting the content and style of the dataset used for fine-tuning.
- Includes actionable steps and might cater to a particular audience's preferences or needs.
- The advice is more structured and could align with expert opinions or best practices.

"To maximize productivity, start by setting clear, achievable goals. Break down your goals into daily tasks and use a digital tool or planner to track your progress. Implement the Pomodoro Technique: work for 25 minutes, then take a 5-minute break. Avoid multitasking, as it can reduce the quality of your work and lead to burnout. Stay hydrated and ensure you get 7-9 hours of sleep each night."



Training an LLM to Follow Instructions

Reinforcement Learning Fine-tuning

Prompt

Output

"How to be productive..."

Refine the model's outputs to align with desired behaviors, ethical guidelines, or qualitative criteria, often based on feedback loops with human evaluators.

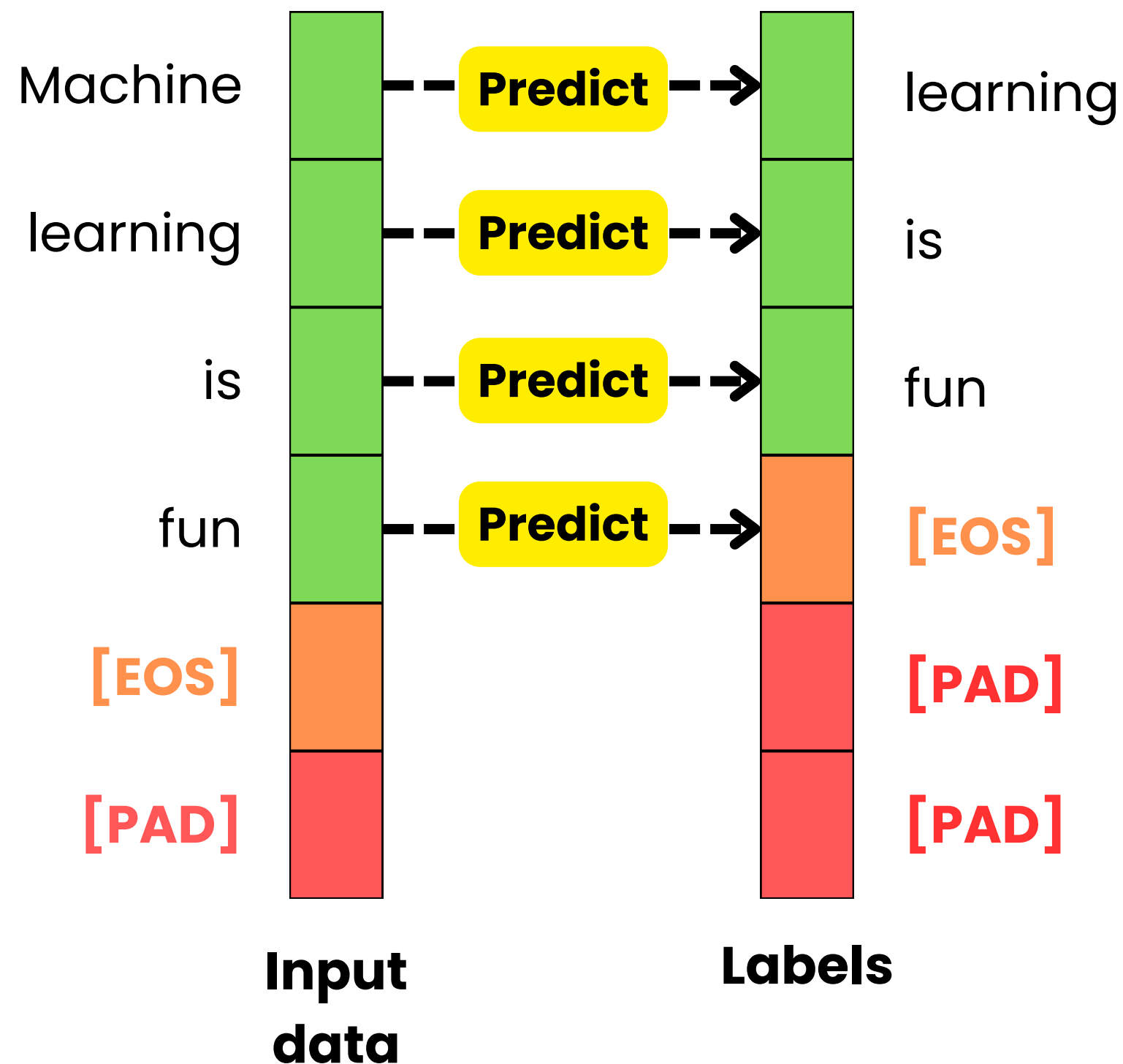
- Ethically aligned and considers the well-being of the user.
- Tailored advice that emphasizes sustainability and personal values.
- Could reflect nuanced human feedback aimed at promoting healthy habits and avoiding overwork.

"To stay productive, first, ensure you're setting realistic goals that align with your long-term objectives. Prioritize tasks that move you closer to these goals and delegate or postpone less critical tasks. Consider using productivity apps to minimize distractions and keep track of deadlines. Remember, productivity isn't about being busy; it's about making meaningful progress on what matters most to you. Finally, make sure to balance hard work with adequate rest to maintain long-term productivity without burnout."



Causal Language Modeling Pretraining

**Self-supervised
learning training**





Causal Language Modeling

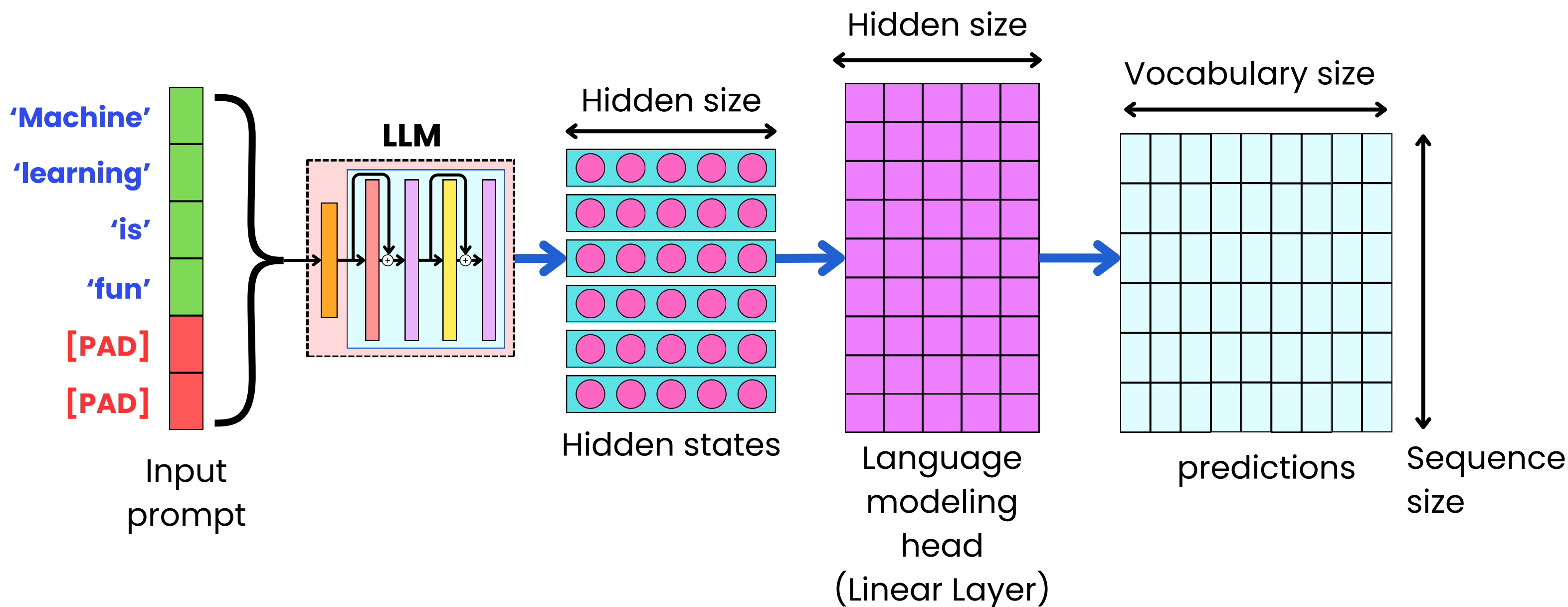
Pretraining

Data To train GPT-3

Dataset	Quantity (tokens)	Weight in training mix	URL
Common Crawl (filtered)	410 billion	60%	https://commoncrawl.org/overview
WebText2	19 billion	22%	https://openwebtext2.readthedocs.io/en/latest/
Books1	12 billion	8%	https://github.com/soskek/bookcorpus
Books2	55 billion	8%	https://github.com/soskek/bookcorpus
Wikipedia	3 billion	3%	https://huggingface.co/datasets/wikipedia

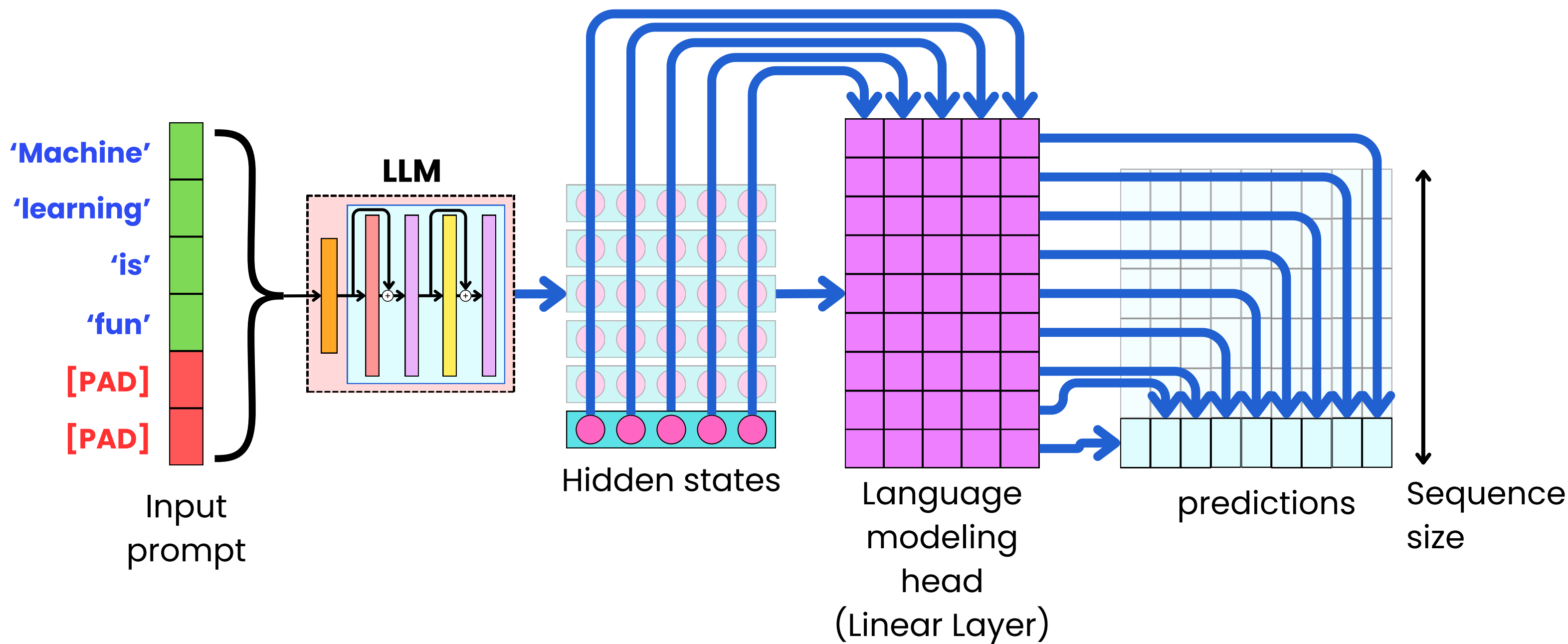


Causal Language Modeling Pretraining



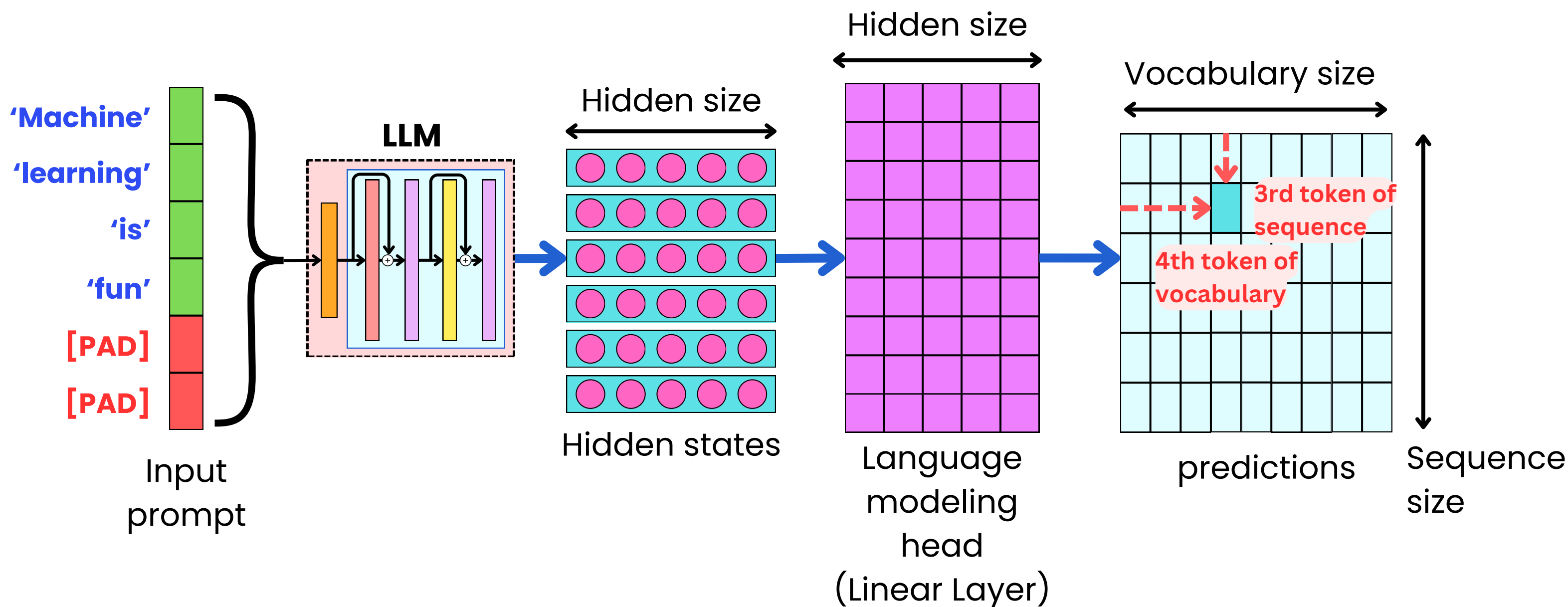


Causal Language Modeling Pretraining



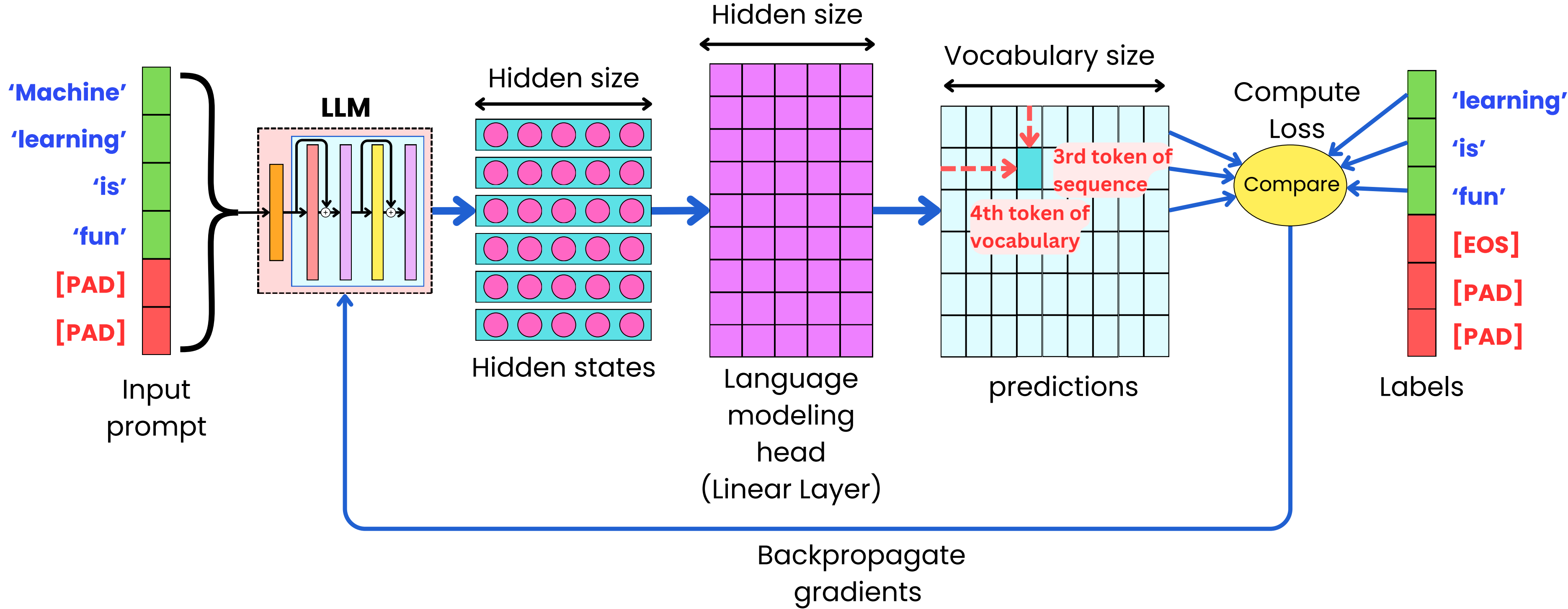


Causal Language Modeling Pretraining





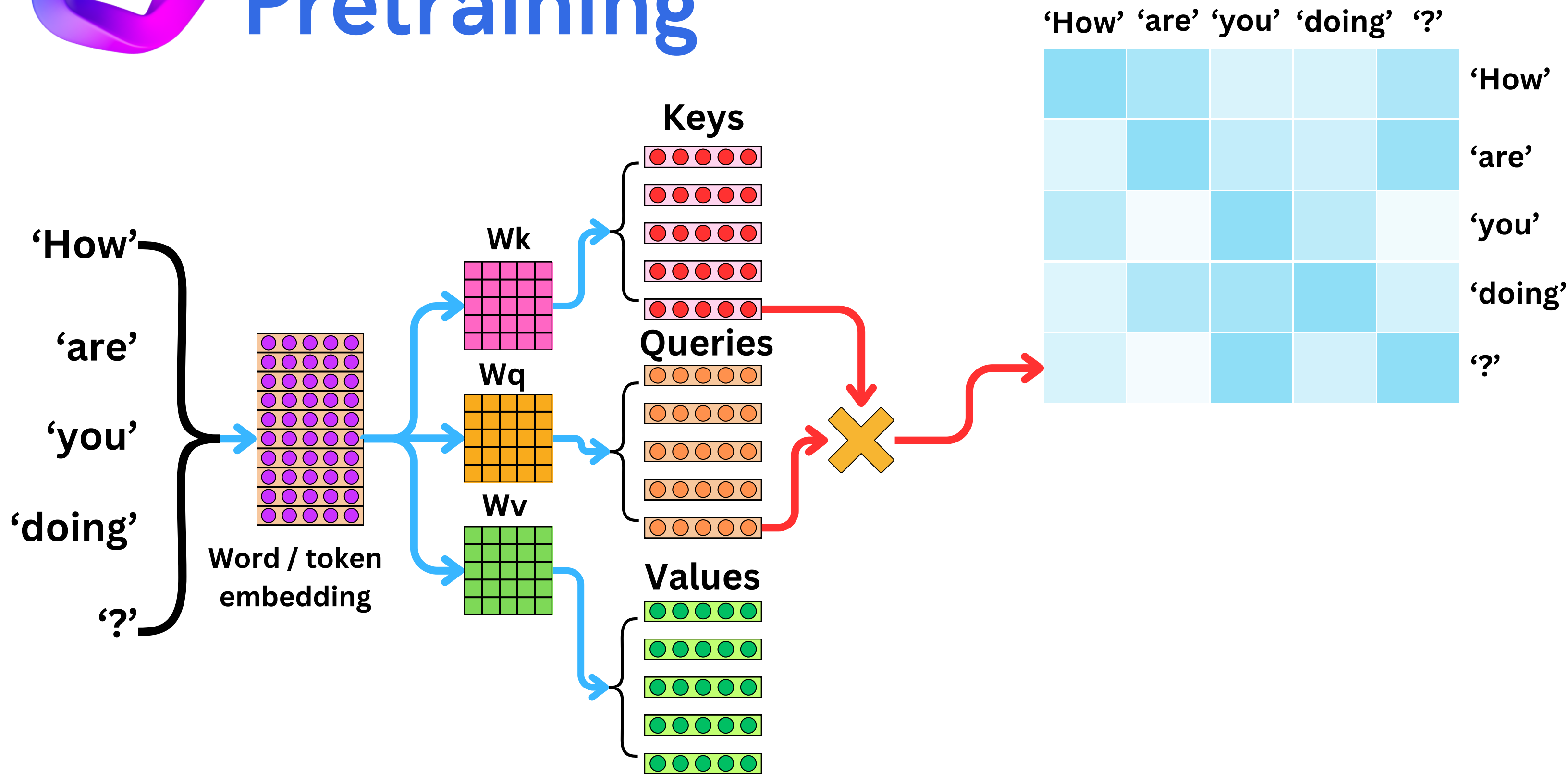
Causal Language Modeling Pretraining





Causal Language Modeling

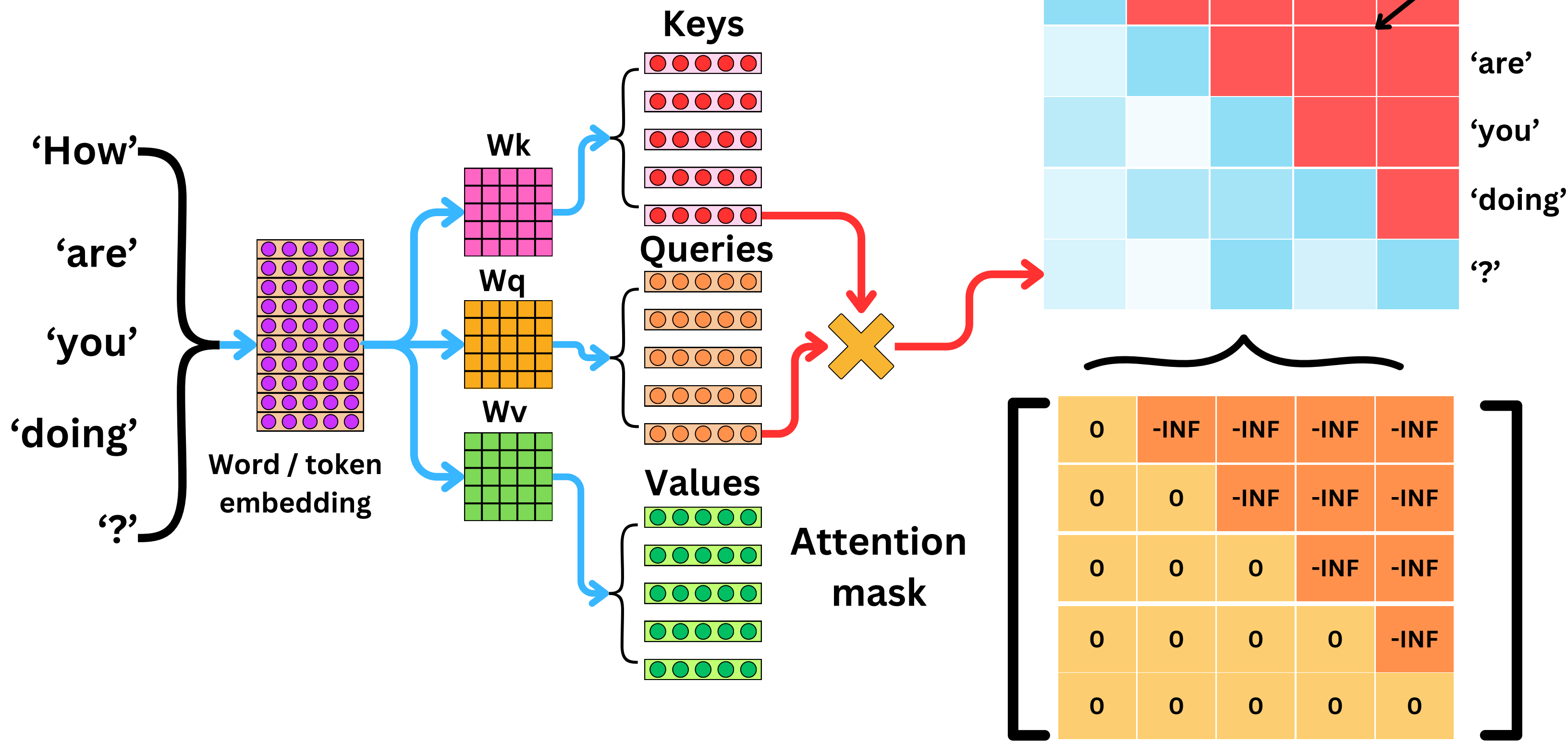
Pretraining





Causal Language Modeling Pretraining

Causal Mask:
attention computed
only for previous
words

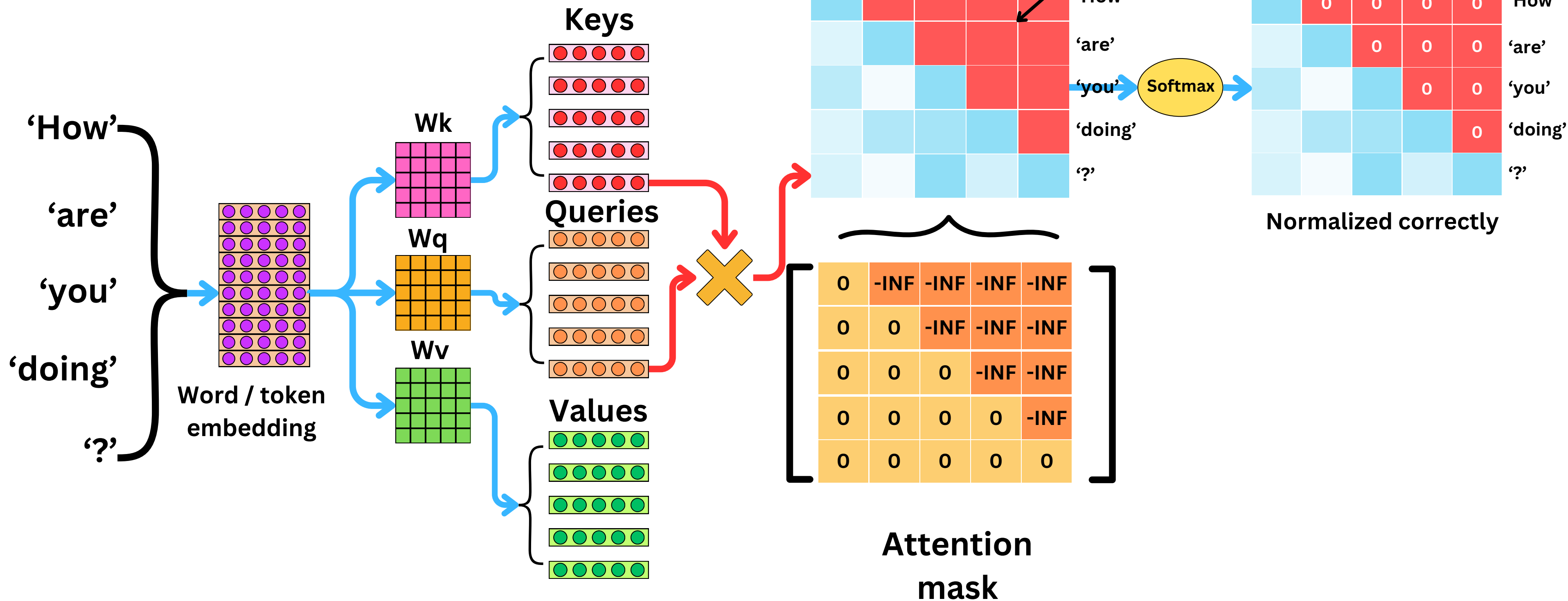




Causal Language Modeling

Causal Mask:
attention computed
only for previous
words

Self-attentions





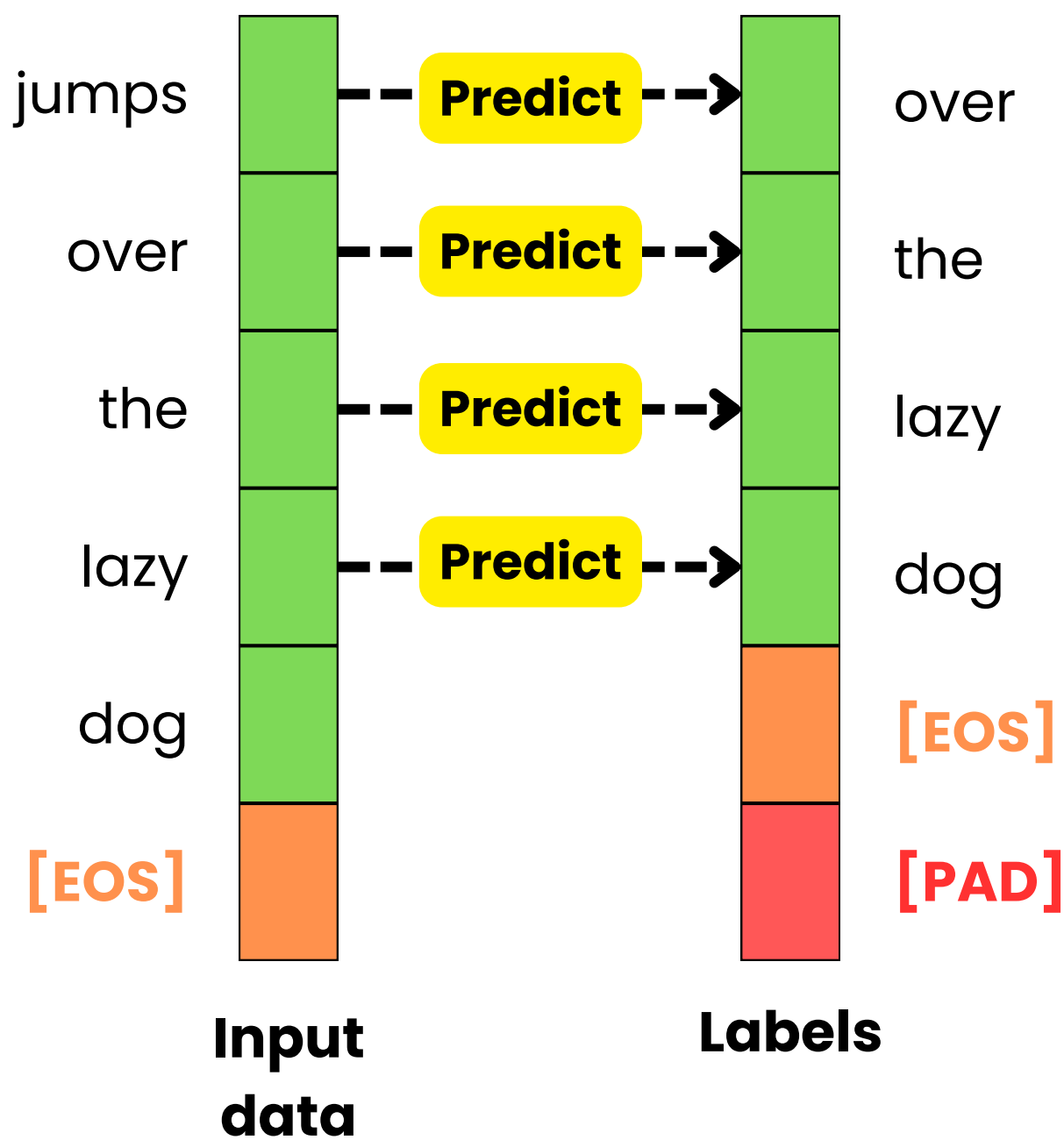
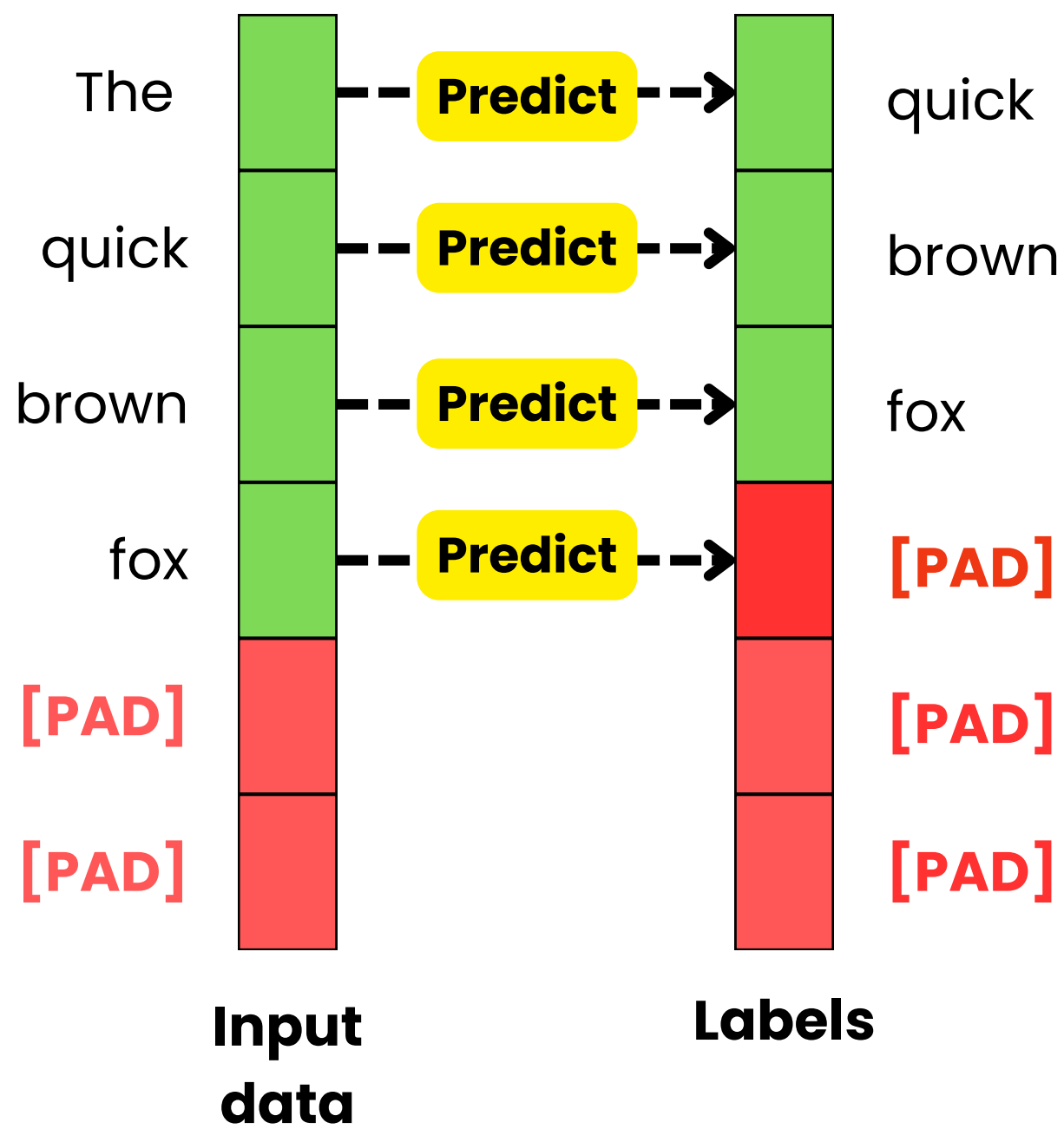
Probability of *ith* token
in Test set

$$\text{Perplexity} = 2^{-\frac{1}{T} \sum_{i=1}^T \log_2 p(t_i)}$$

Total number of tokens
in Test set

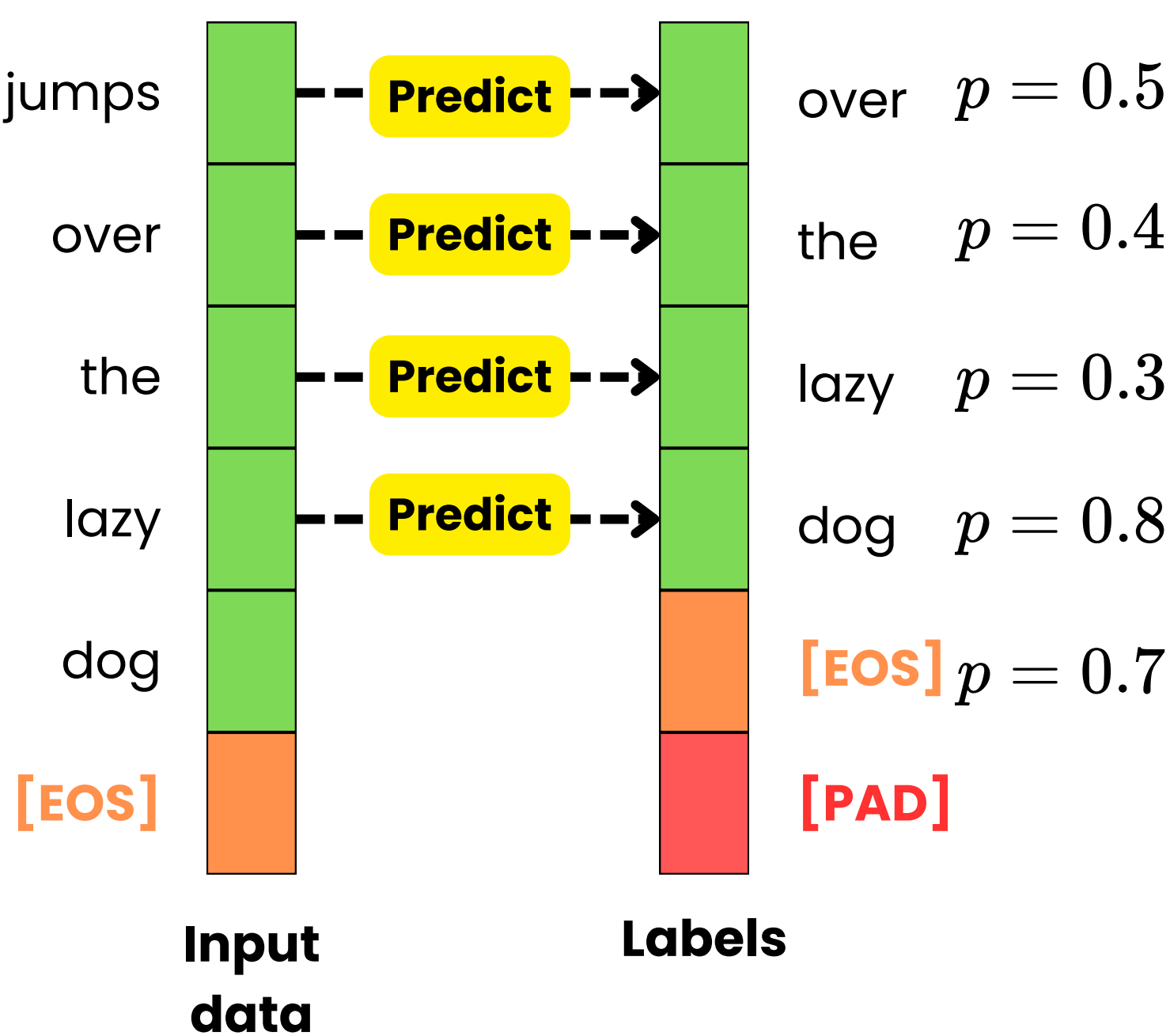
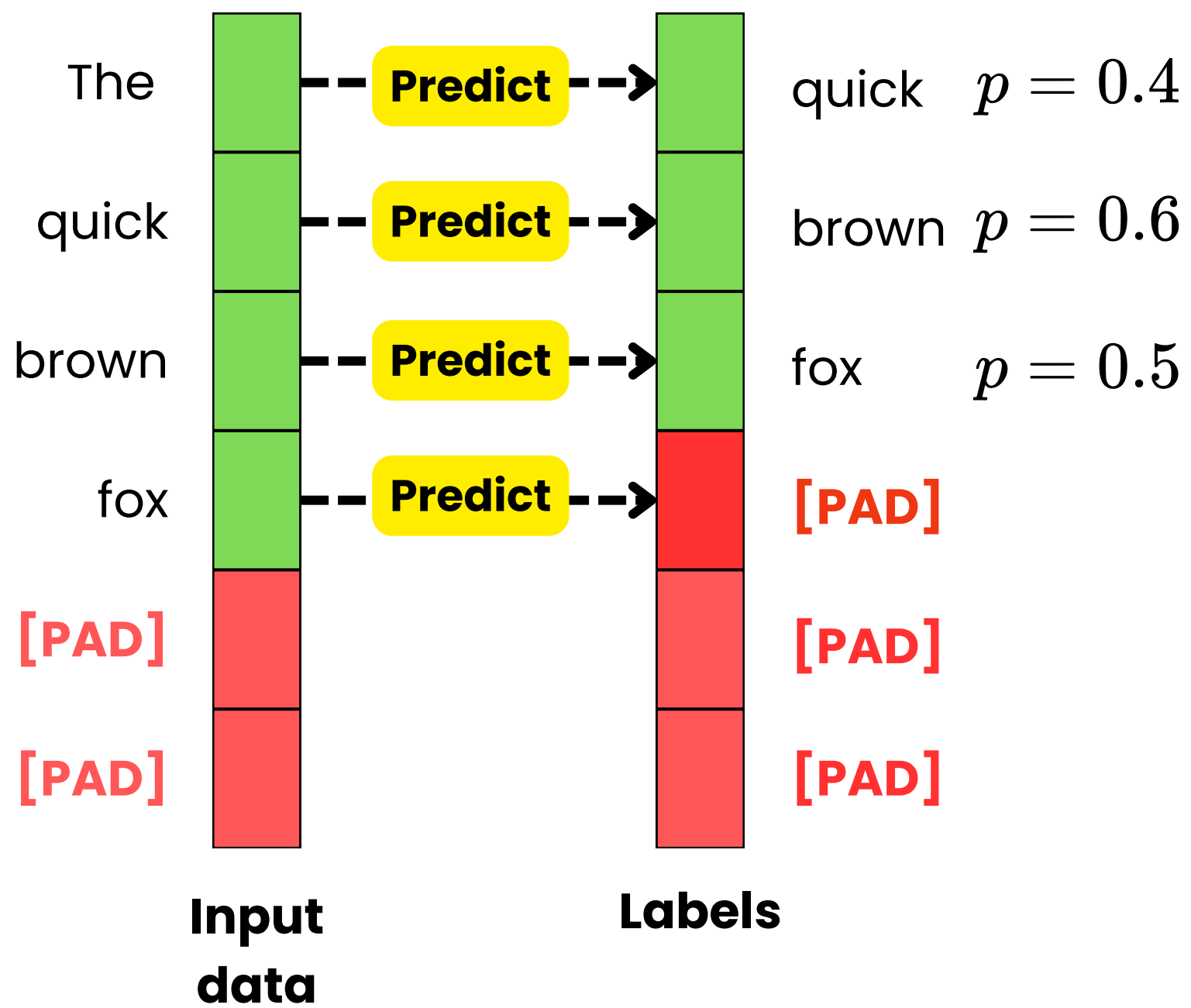


Perplexity





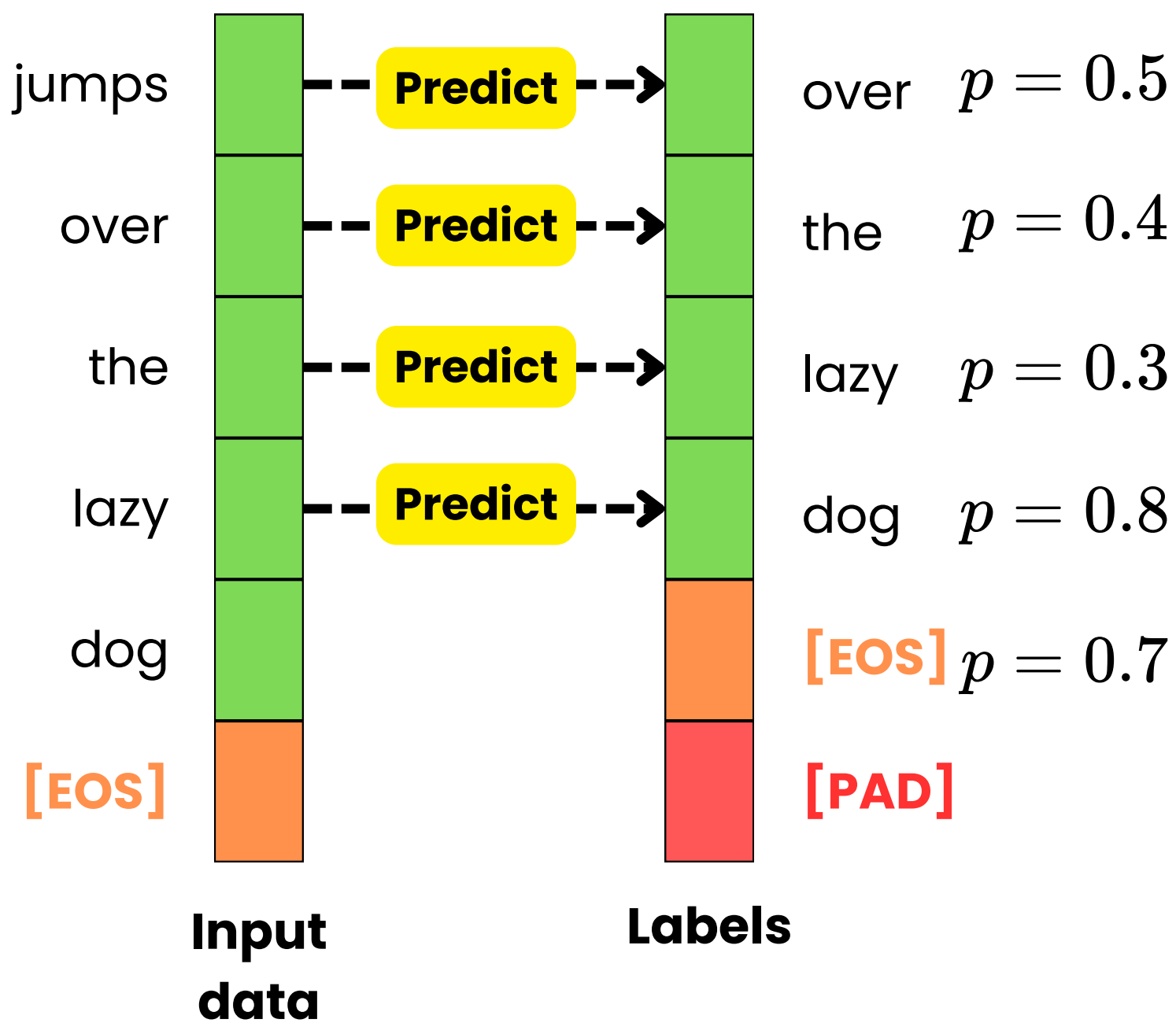
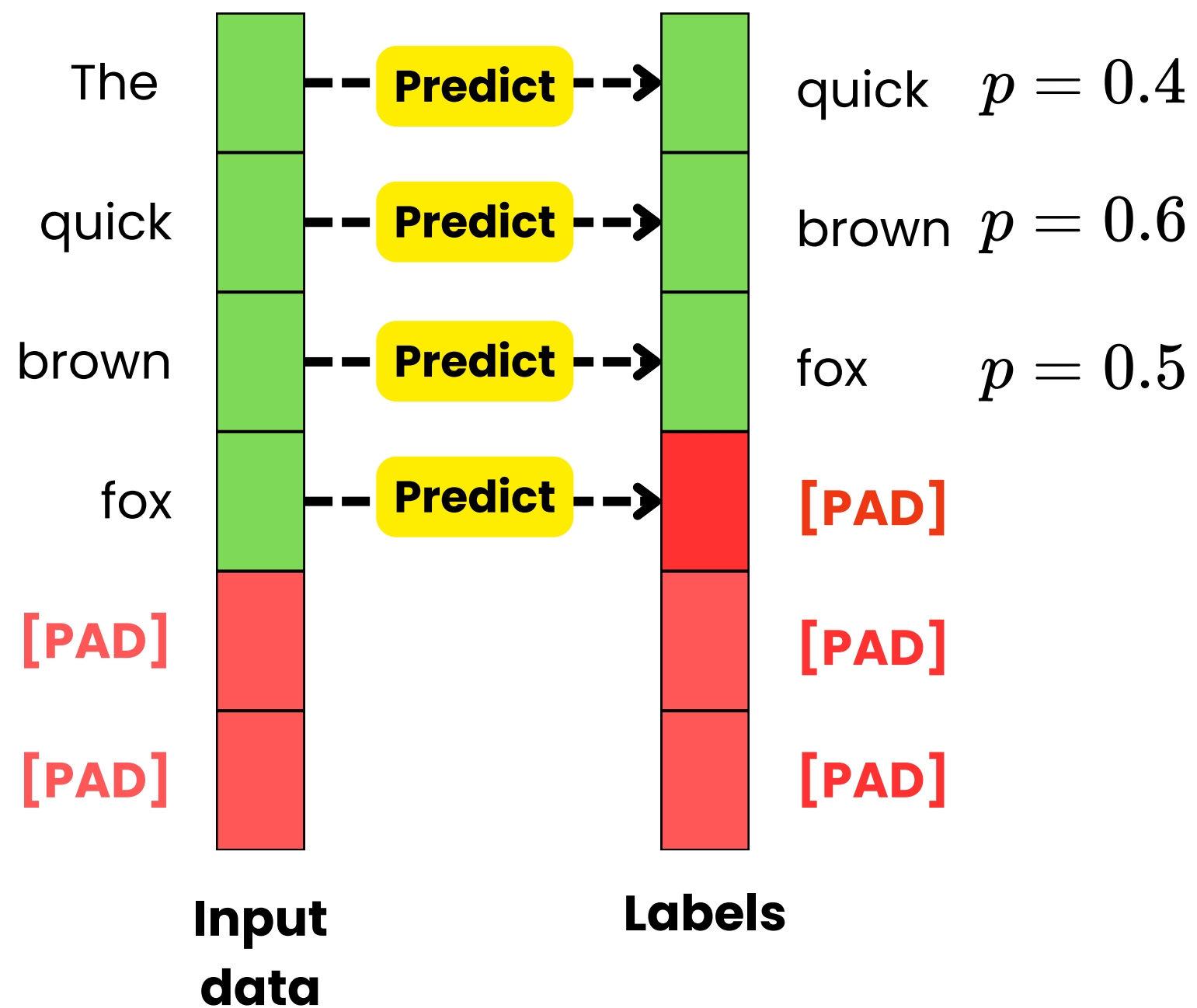
Perplexity





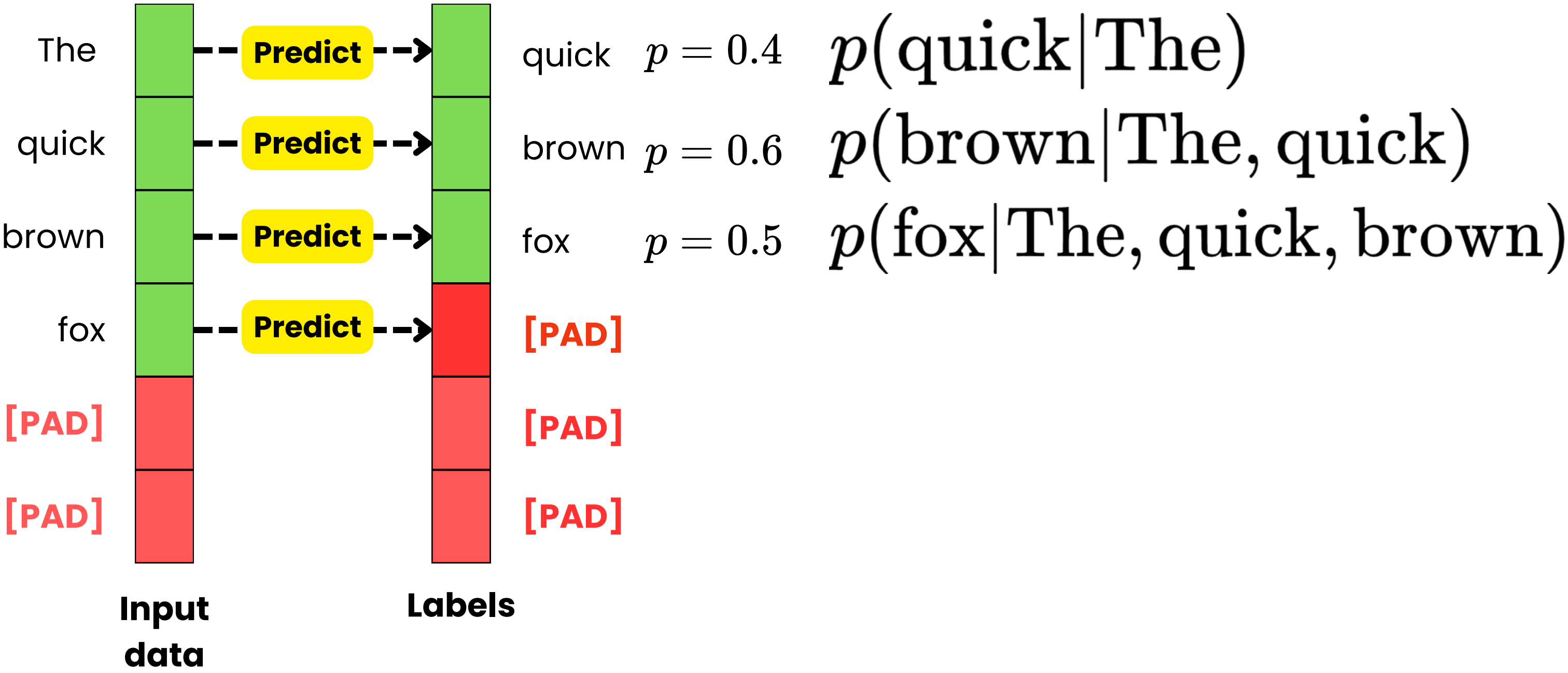
Perplexity

$$\text{Perplexity} = 2^{-\log_2(0.4 \times 0.6 \times 0.5 \times 0.5 \times 0.4 \times 0.3 \times 0.8 \times 0.7) / 8}$$



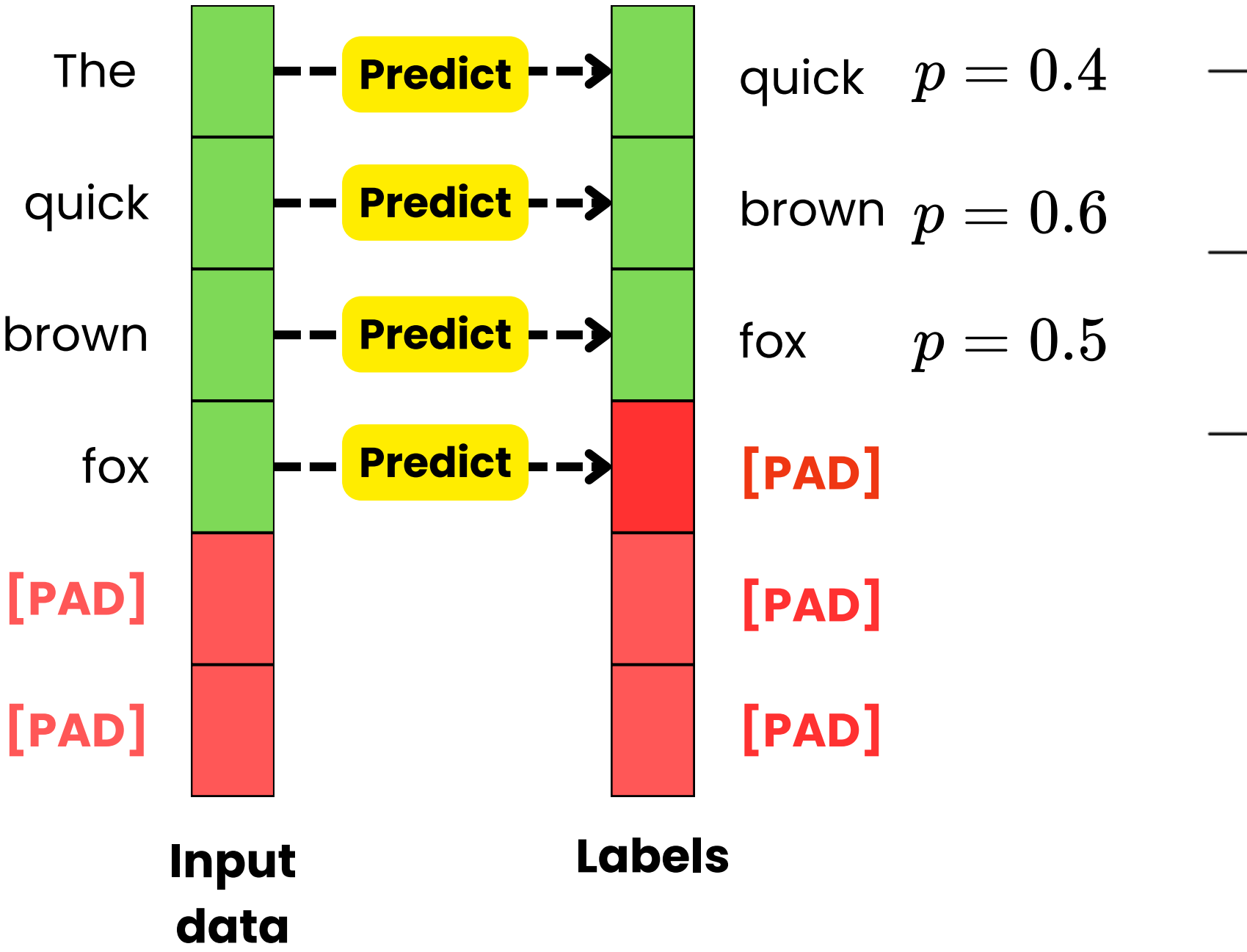


Perplexity





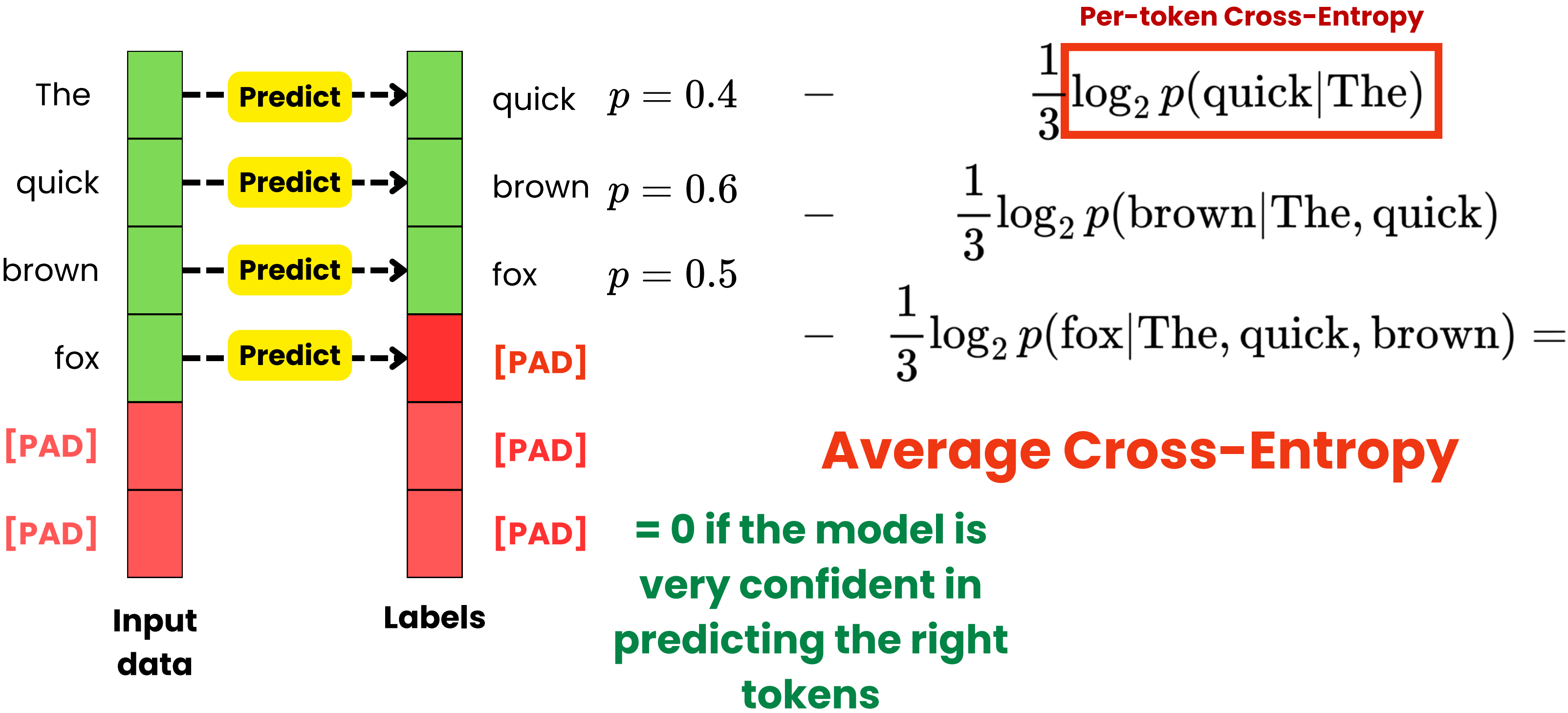
Perplexity



$$\begin{aligned} & - \frac{1}{3} \log_2 p(\text{quick} | \text{The}) \\ & - \frac{1}{3} \log_2 p(\text{brown} | \text{The, quick}) \\ & - \frac{1}{3} \log_2 p(\text{fox} | \text{The, quick, brown}) = \end{aligned}$$

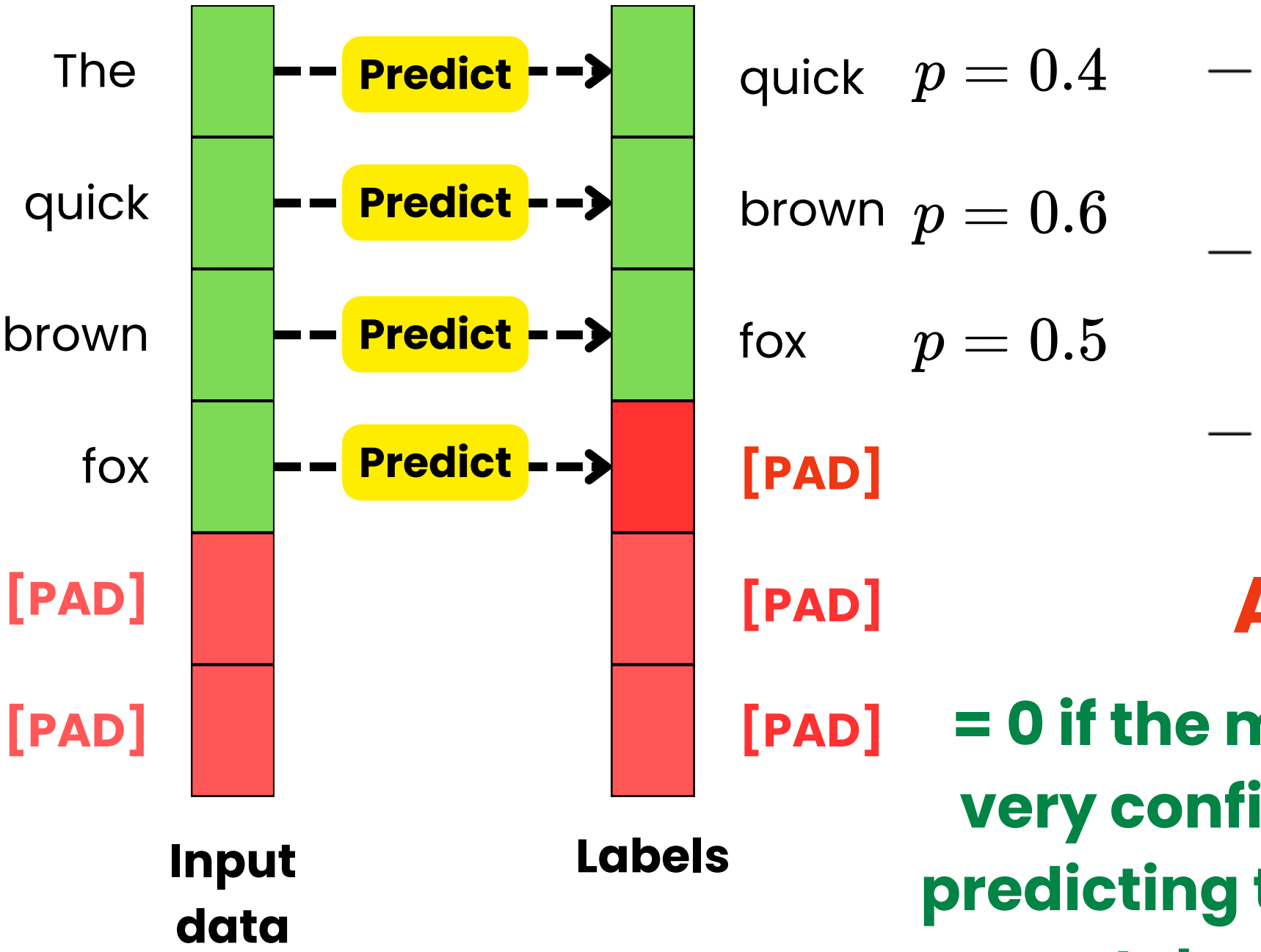


Perplexity





Perplexity



Per-token Cross-Entropy

$$\frac{1}{3} \log_2 p(\text{quick} | \text{The})$$

$$\frac{1}{3} \log_2 p(\text{brown} | \text{The, quick})$$

$$\frac{1}{3} \log_2 p(\text{fox} | \text{The, quick, brown}) =$$

Average Cross-Entropy

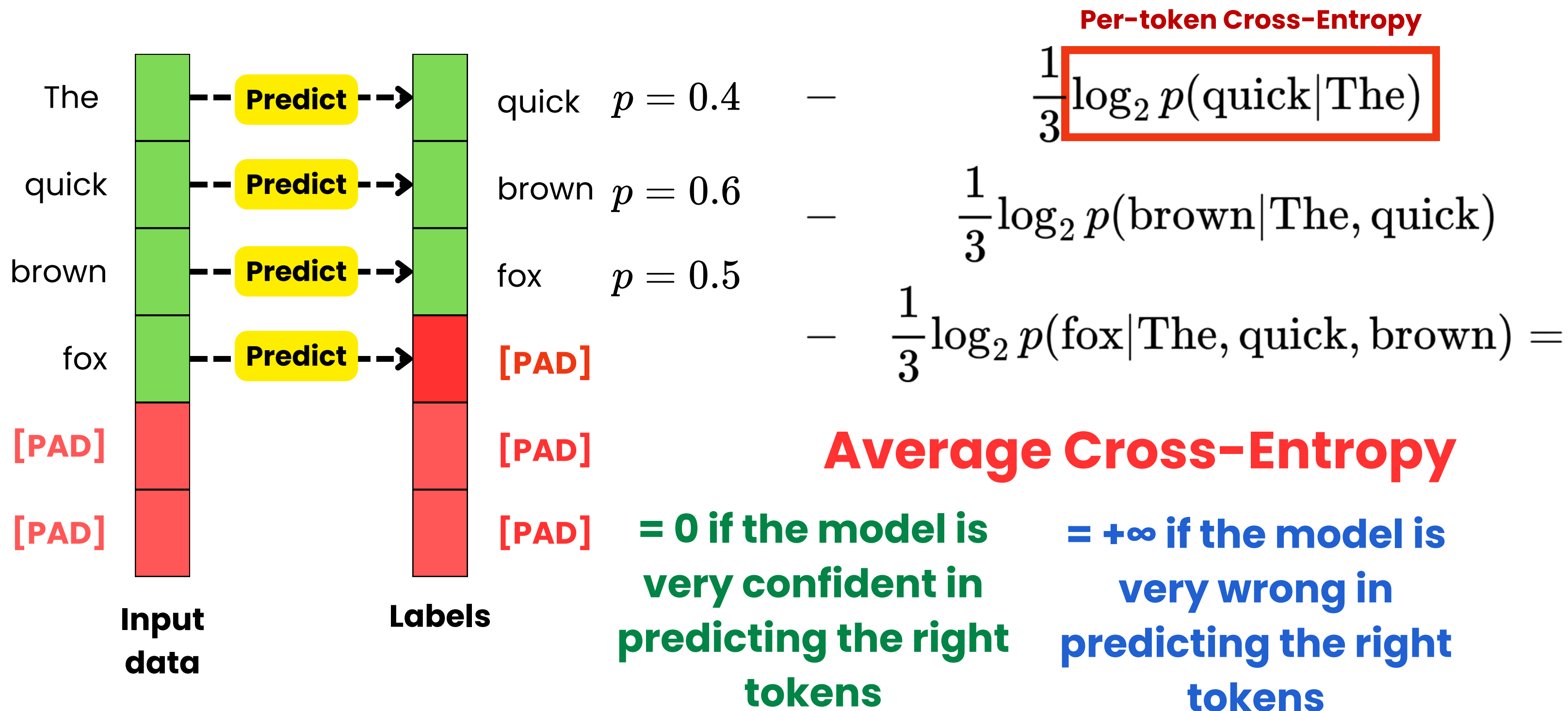
= 0 if the model is
very confident in
predicting the right
tokens

= $+\infty$ if the model is
very wrong in
predicting the right
tokens



Perplexity

So a good model minimizes **Average Cross-Entropy**





Perplexity

$$\text{Perplexity} = 2^{\text{Average Cross-Entropy}}$$

**= 1 if the model is
very confident in
predicting the right
tokens**

**= $+\infty$ if the model is
very wrong in
predicting the right
tokens**

**Perplexity tells us how confused is the
model in predicting tokens:**

$\Rightarrow$ We want to a model that is not confused

$\Rightarrow$ We want to a model that is certain



Perplexity

$$\text{Perplexity} = 2^{\text{Average Cross-Entropy}}$$

- A perplexity of 100 suggests that for each word, the model is as uncertain as if it were choosing between 100 equally likely options.
- A perplexity of 10 indicates the model is as uncertain as if it were choosing between 10 equally likely options for each word.



Supervised Learning Fine-Tuning

The Data

prompt:
"What is the
capital of
France?"



answer:
"The capital
of France is
Paris."

Pairs of
Prompts - answers



Supervised Learning Fine-Tuning

The Data

Use-case	Prompt
Brainstorming	List five ideas for how to regain enthusiasm for my career
Generation	Write a short story where a bear goes to the beach, makes friends with a seal, and then returns home.
Rewrite	This is the summary of a Broadway play: {summary} This is the outline of the commercial for that play:



Supervised Learning Fine-Tuning

Example of Data

Stanford
Alpaca



<https://crfm.stanford.edu/2023/03/13/alpaca.html>

52K instruction-following demonstrations

Below is an instruction that describes a task. Write a response that appropriately completes the request.

Instruction:

Develop a plan to reduce electricity usage in a home.

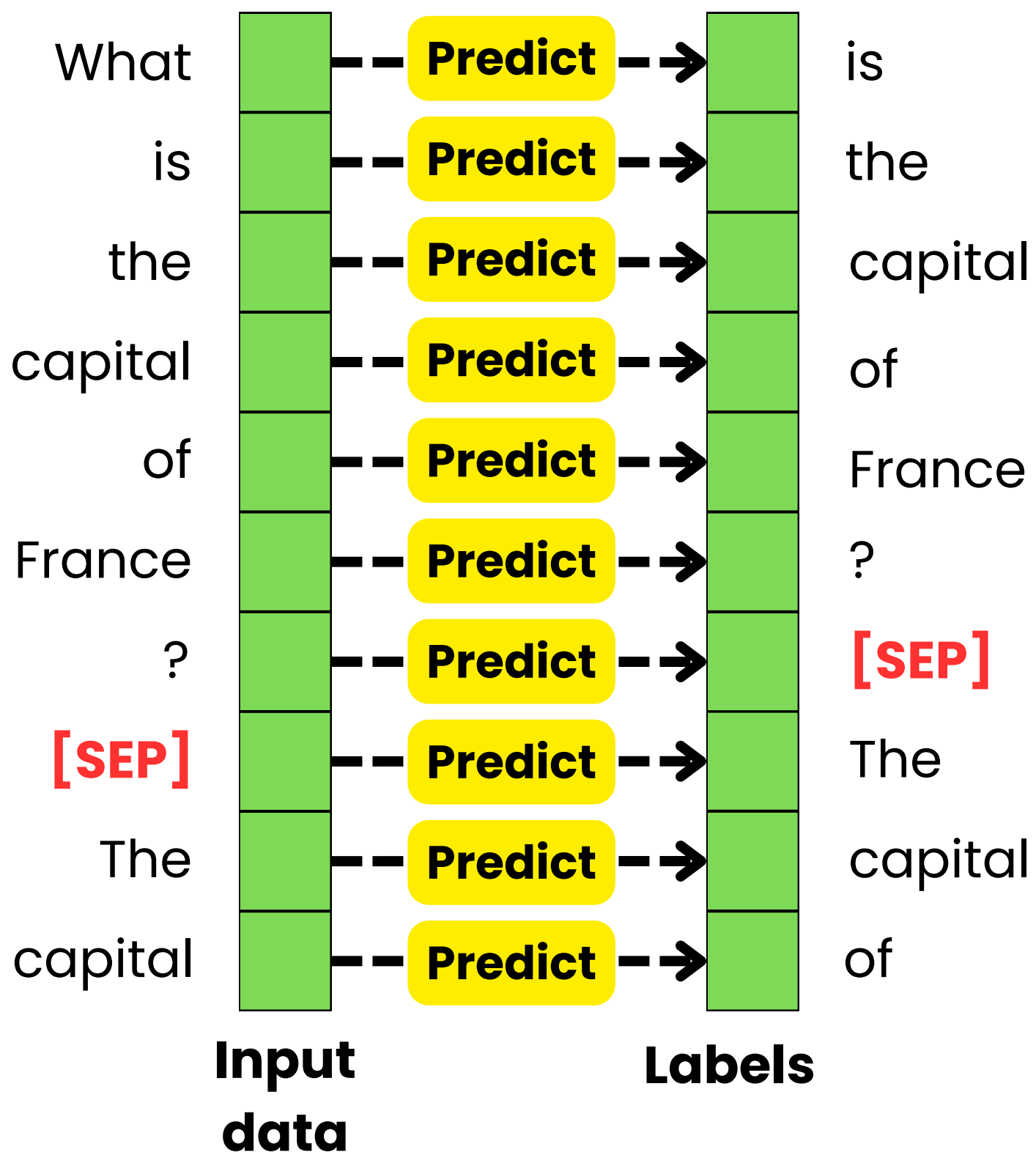
Response:

A plan to reduce electricity usage in a home can include a number of different strategies. These include replacing inefficient lightbulbs with energy-efficient LEDs, unplugging appliances and electronics when not in use, using natural light instead of electric lights where possible, using a touch-free faucet to reduce water heater use, using a smart thermostat to monitor and adjust temperatures automatically, installing a solar hot water heater, and turning off electronics when not in use. Additionally, making sure to install and seal windows and doors correctly to reduce air leakage can also help reduce electricity usage.



Supervised Learning Fine-Tuning

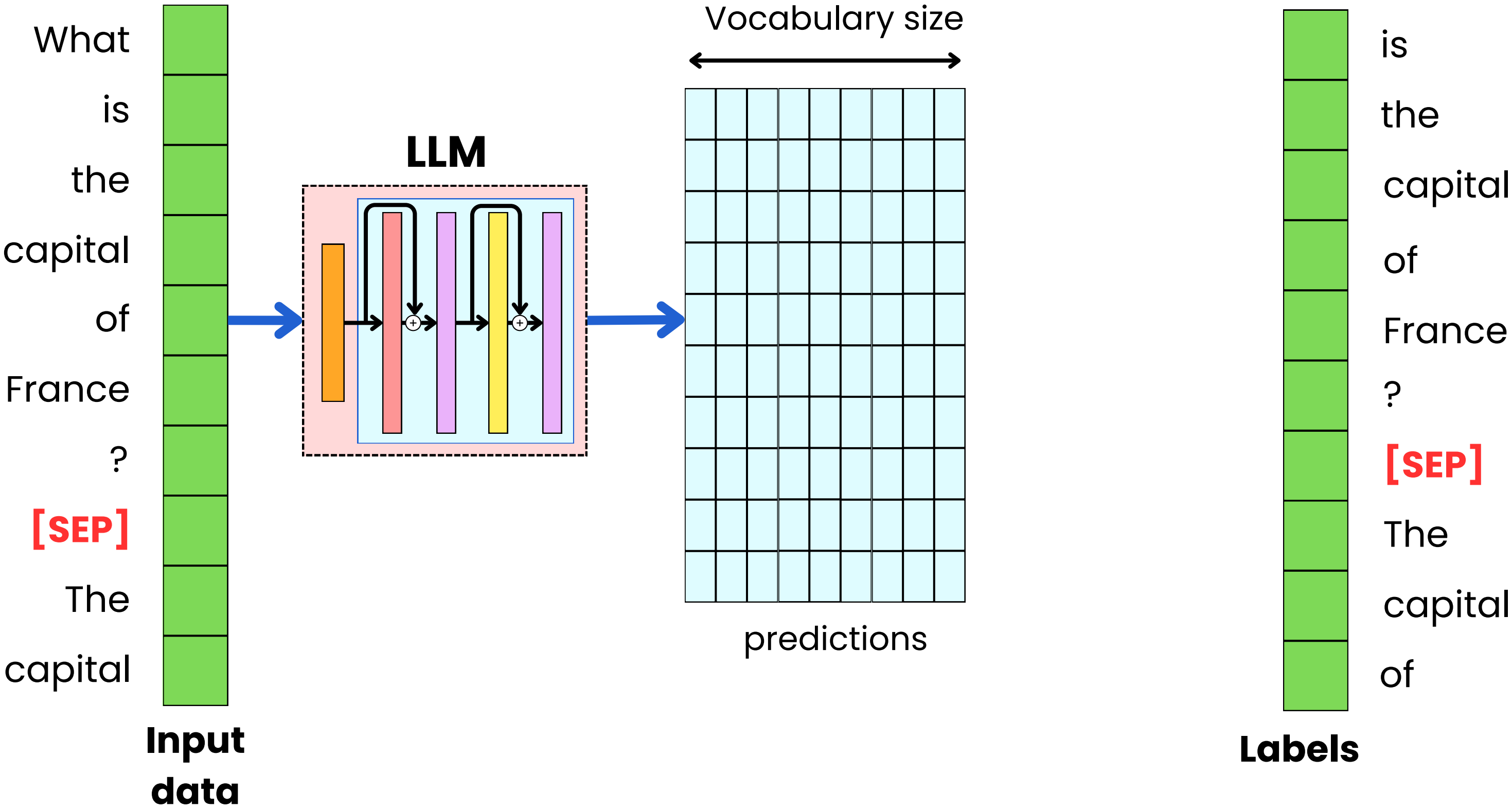
Autoregressing training





Supervised Learning Fine-Tuning

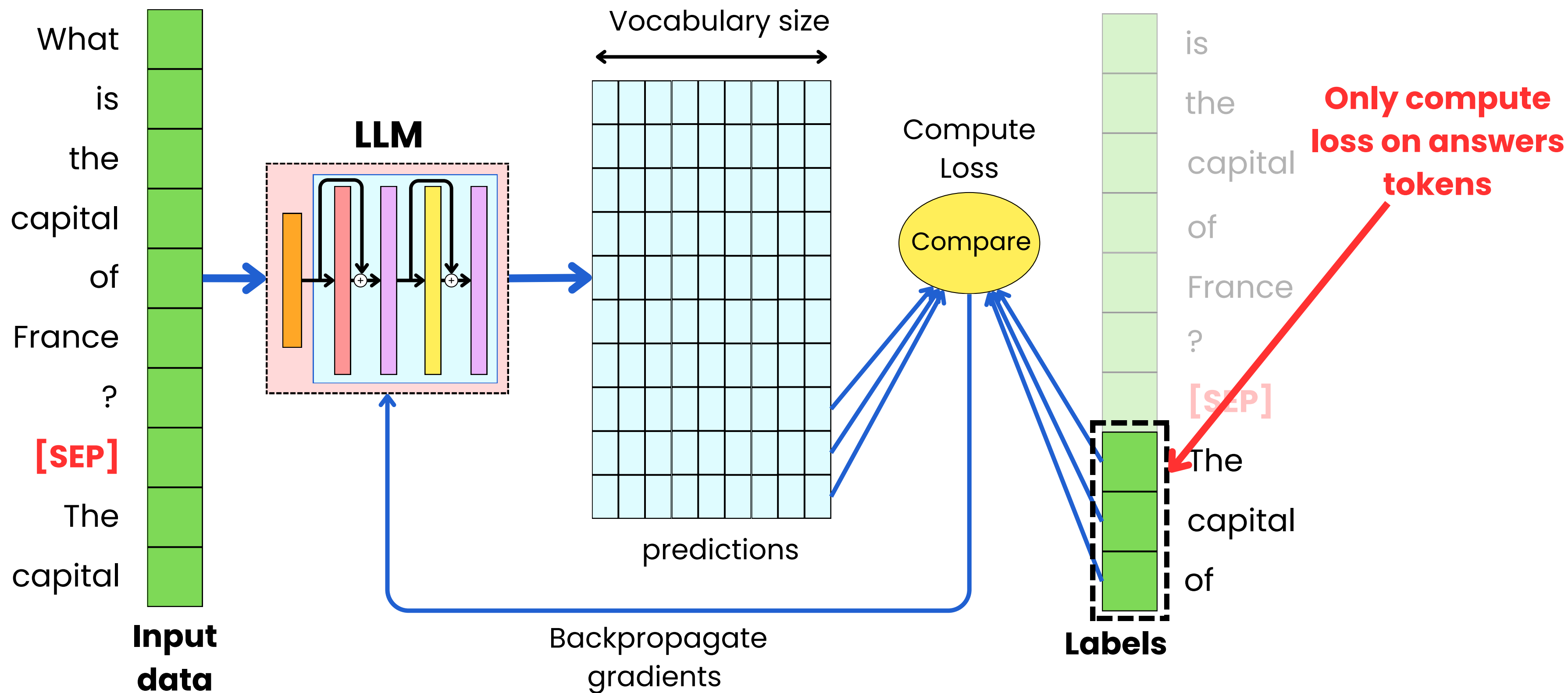
Masking the question tokens





Supervised Learning Fine-Tuning

Masking the question tokens



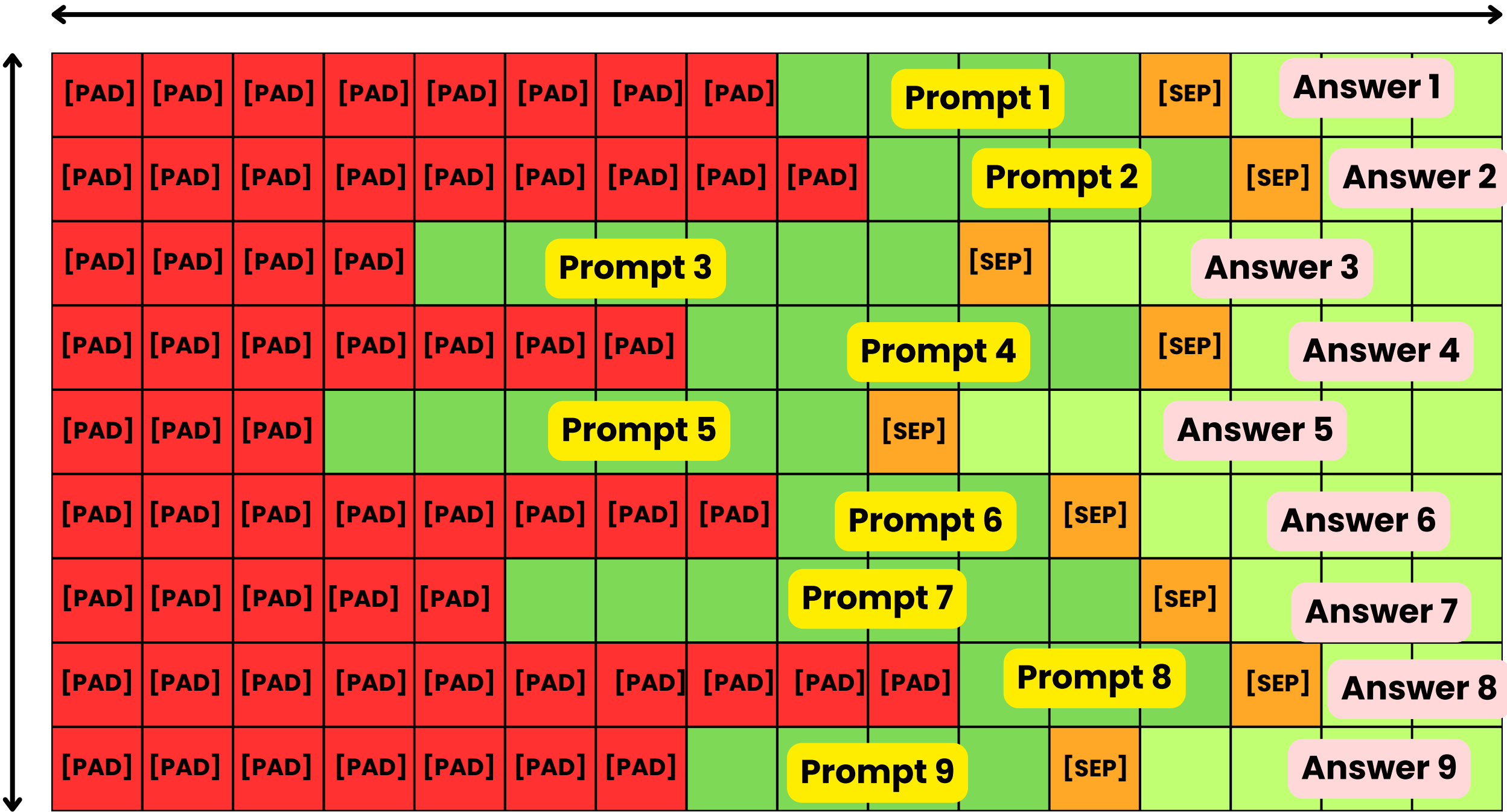


Supervised Learning Fine-Tuning

Preparing the Batches

Maximum number of tokens

Number of
samples in
the batch



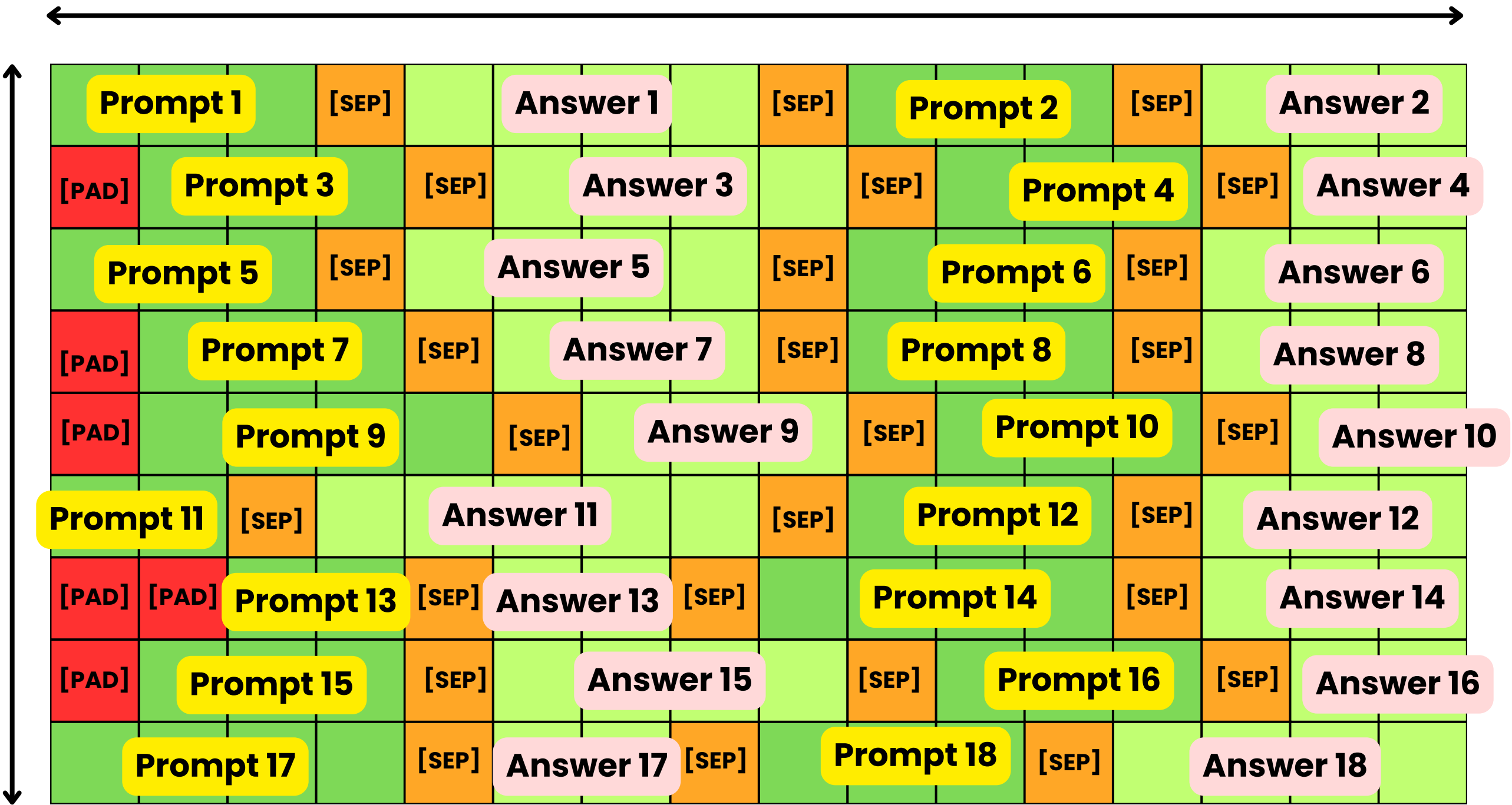


Supervised Learning Fine-Tuning

Packing the Batches

Maximum number of tokens

Number of samples in the batch





Supervised Learning Fine-Tuning

Evaluation: Tasks specific metrics

Task	Benchmark	Metrics
Code	HumanEval, MBPP	pass@k, accuracy
Commonsense Reasoning	PIQA, SIQA, HellaSwag, WinoGrande, ARC easy and challenge, OpenBookQA, CommonsenseQA	accuracy
World Knowledge	NaturalQuestions, TriviaQA	F1 score and exact match (EM)
Reading Comprehension	SQuAD, QuAC, BoolQ	F1 score, exact match, accuracy
MATH	GSM8K, MATH	accuracy



Supervised Learning Fine-Tuning

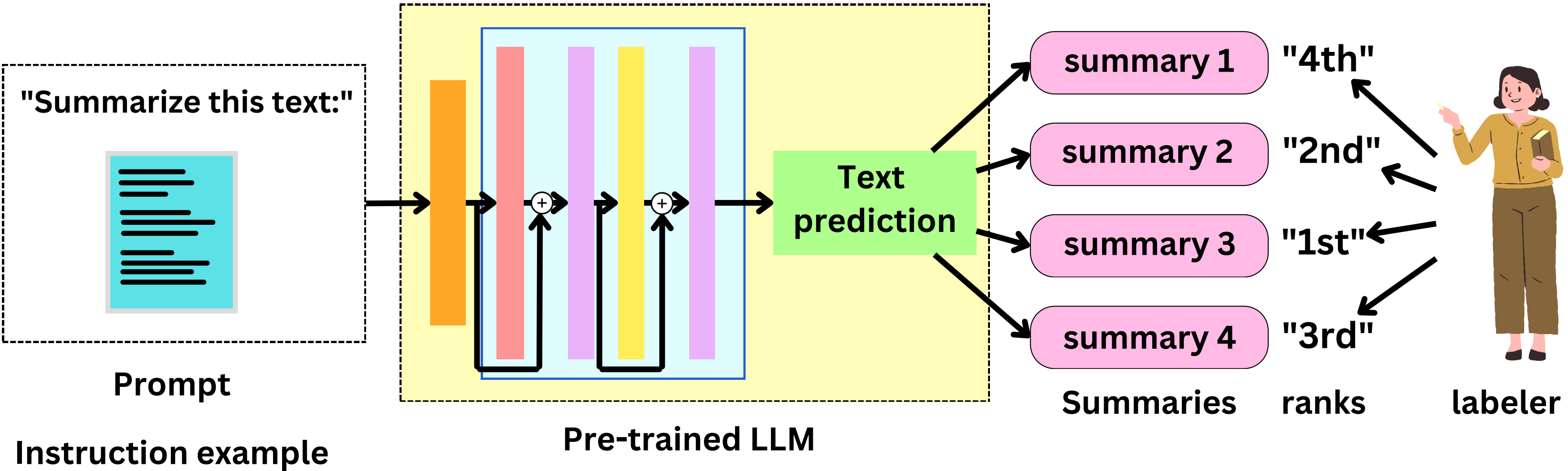
Evaluation: Tasks specific metrics

Task	Benchmark	Metrics
General language understanding across a broad range of subjects and disciplines	MMLU (Massive Multitask Language Understanding)	accuracy
Advanced reasoning, creativity, and understanding nuanced or abstract concepts	Big Bench Hard	F1 score, accuracy
Broad spectrum of cognitive abilities	AGI Eval	Accuracy, F1 score, completion rate



Reinforcement Learning with Human Feedback (RLHF)

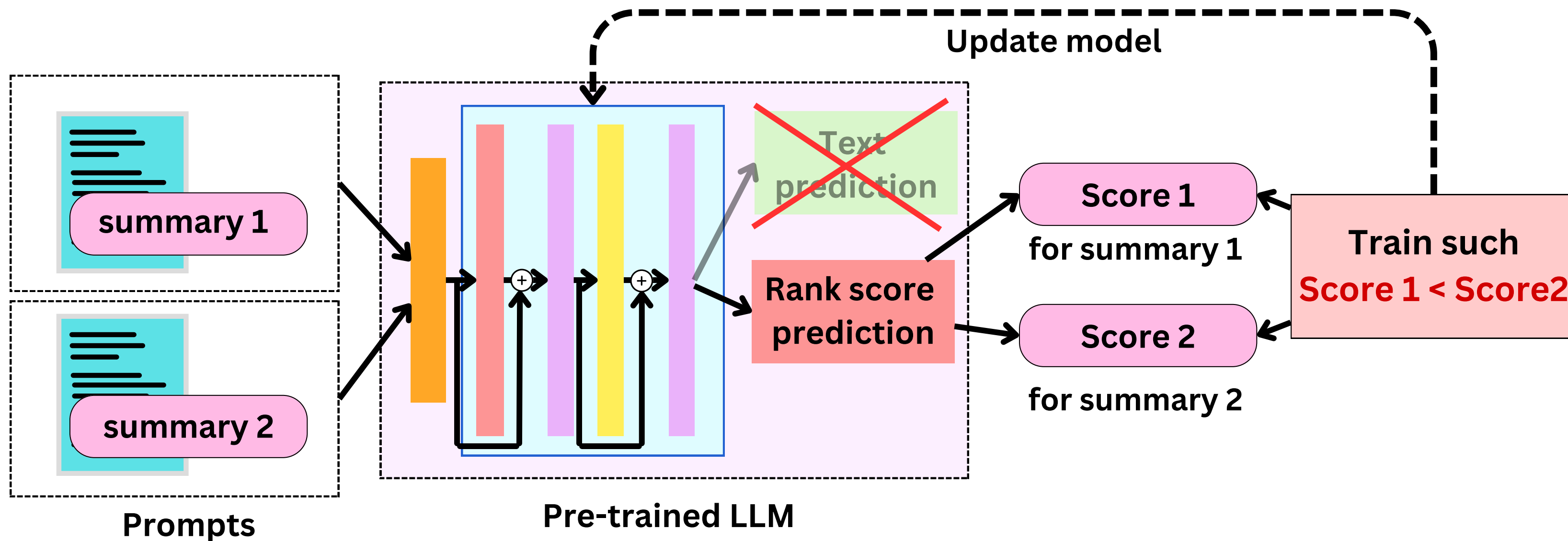
Rank model outputs with human labeler





Reinforcement Learning with Human Feedback (RLHF)

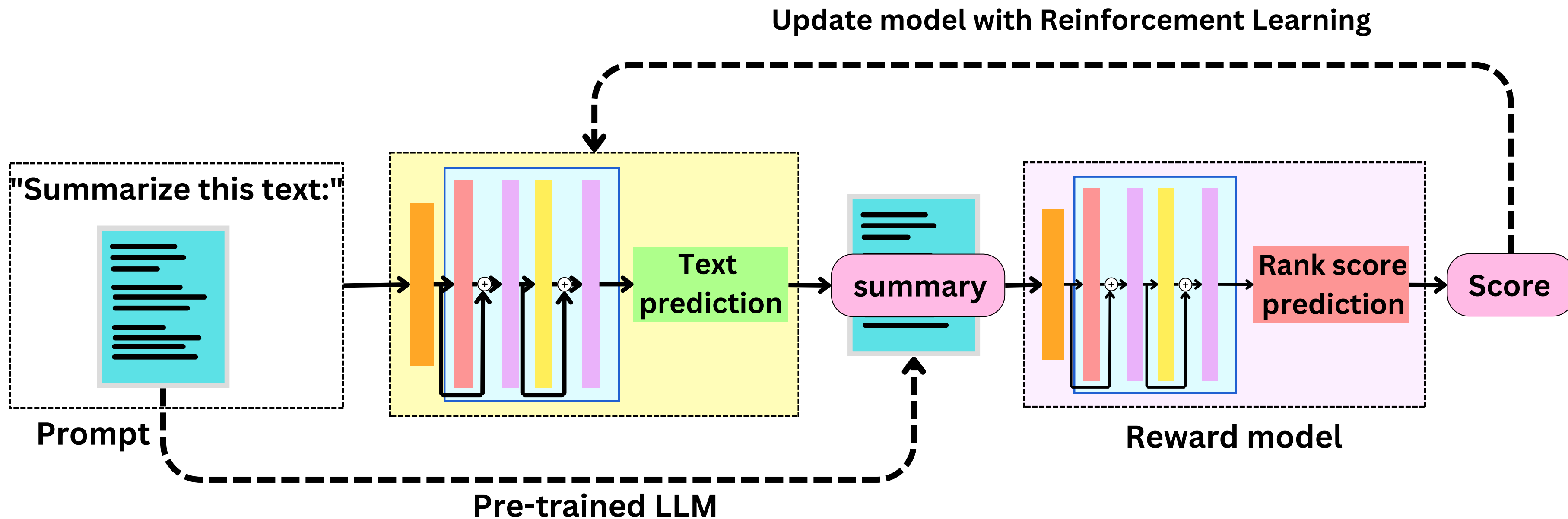
Step 2: Train Reward model to learn to rank output





Reinforcement Learning with Human Feedback (RLHF)

Step 3: Use Reward model to update model with RL



The Data

 Anthropic
`/hh-rlhf`

 `huggingface.co`

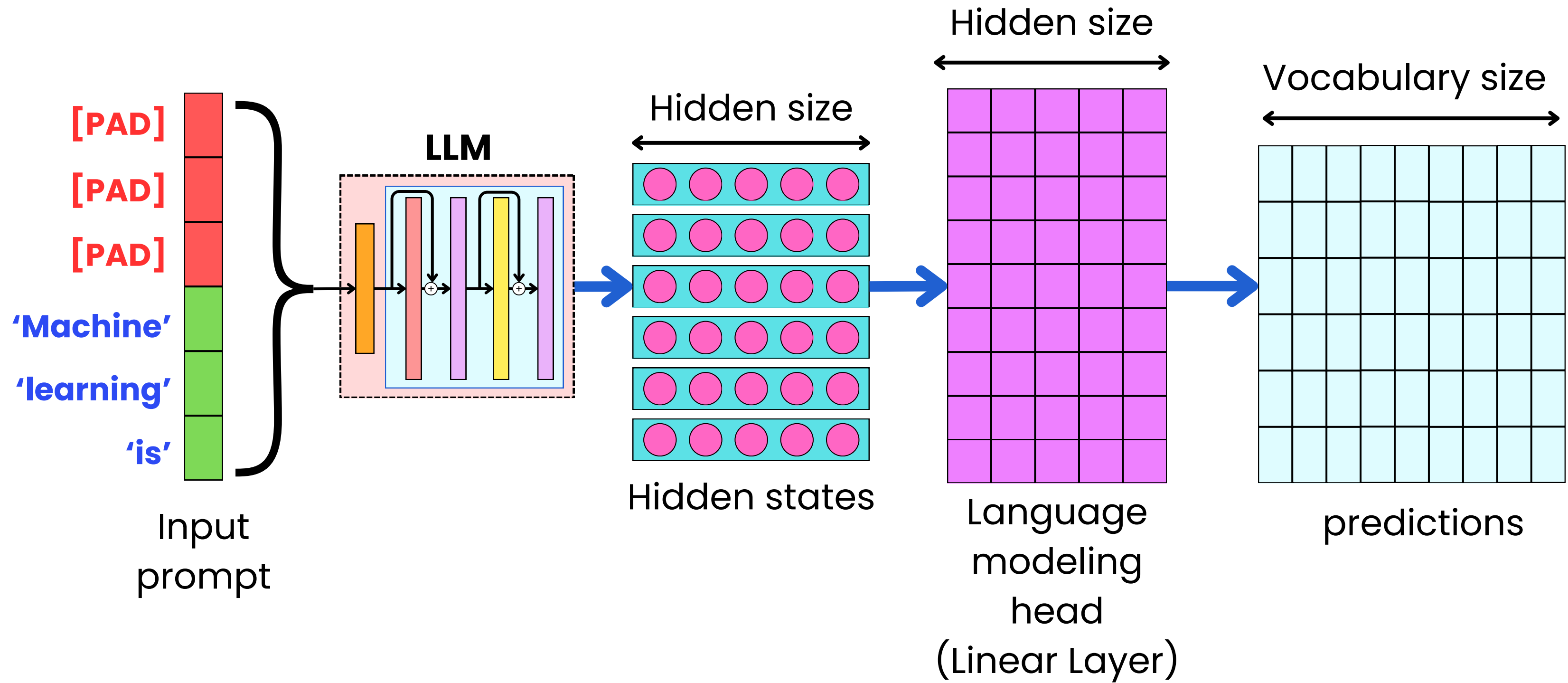
Anthropic/hh-rlhf · Datasets at Hugging Face

We're on a journey to advance and democratize artificial intelligence through open source and open science.

 `huggingface`

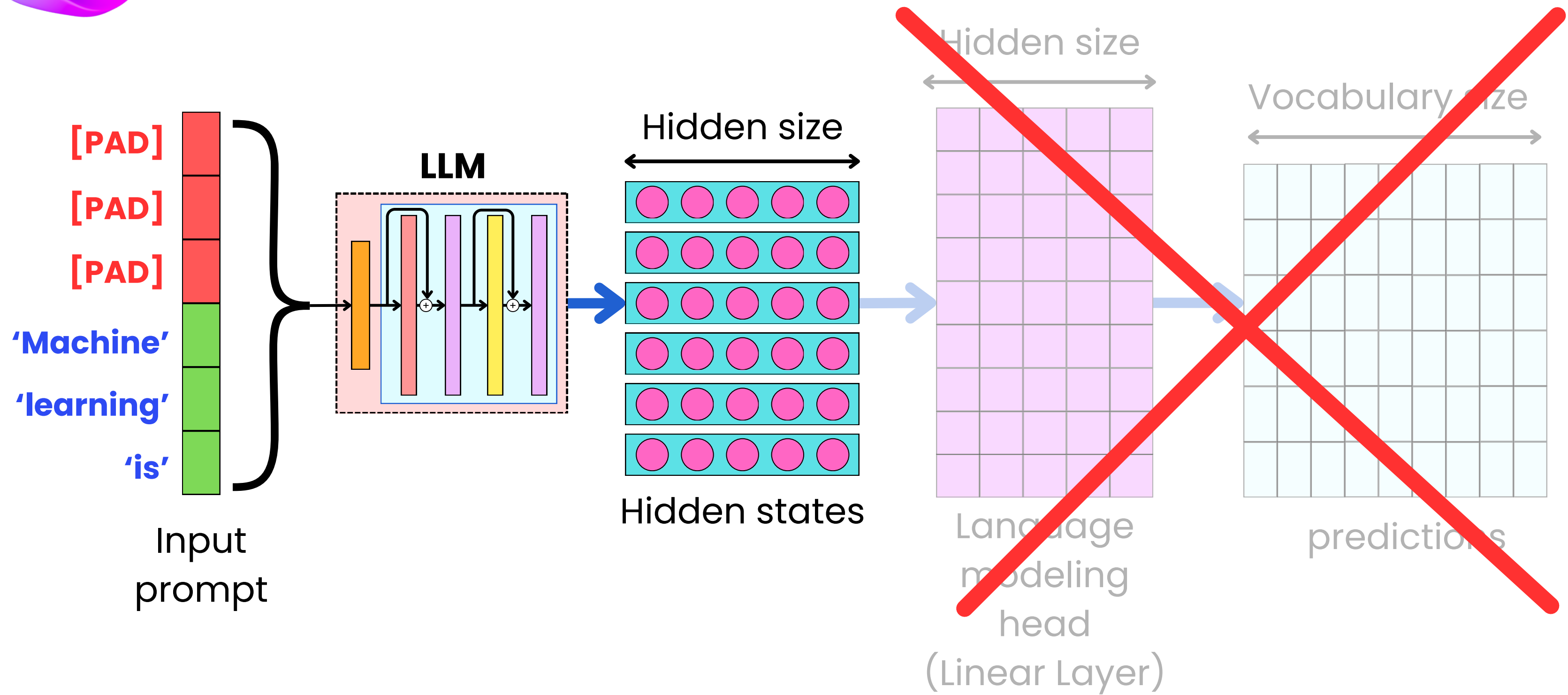


The Reward Model



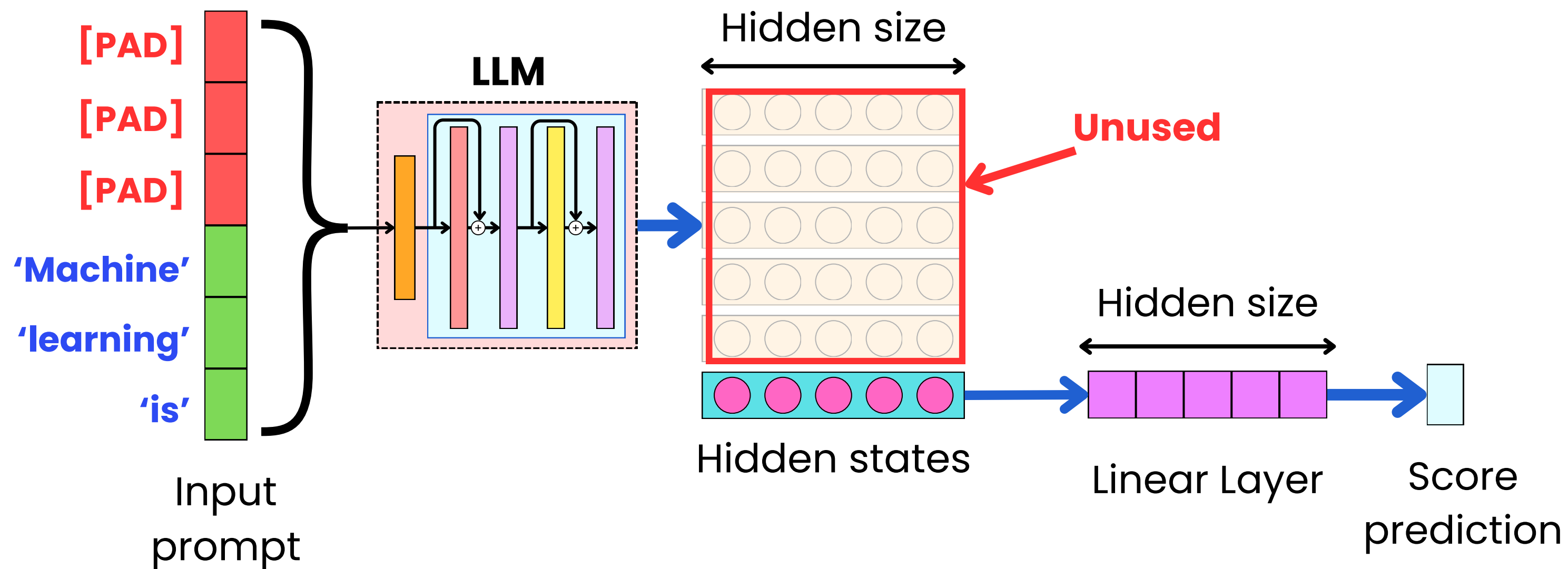


The Reward Model





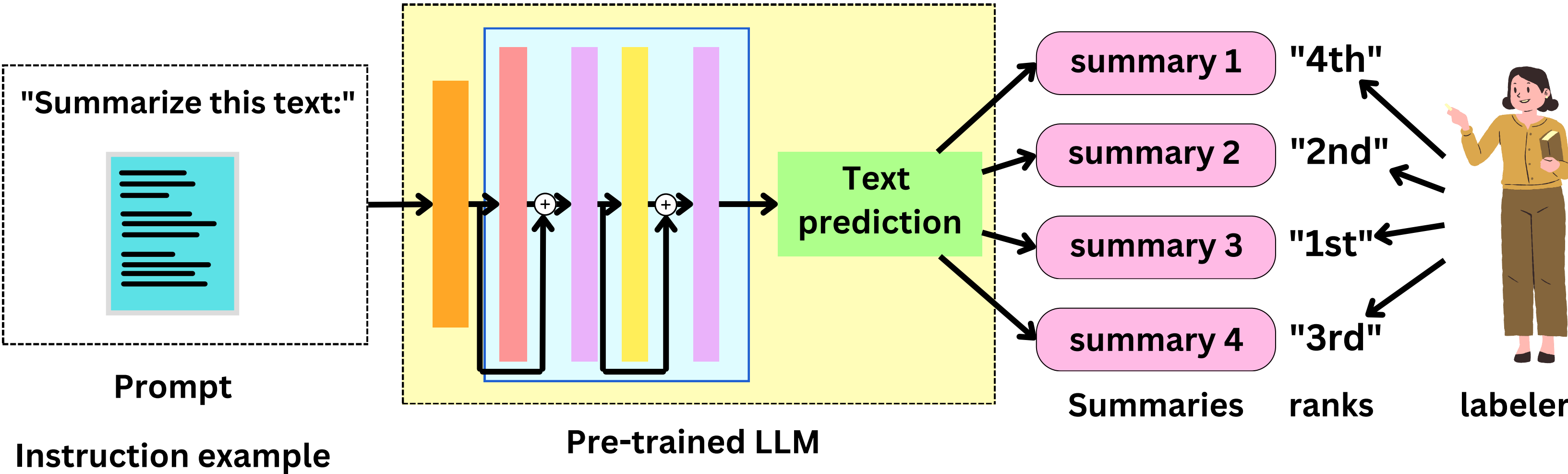
The Reward Model





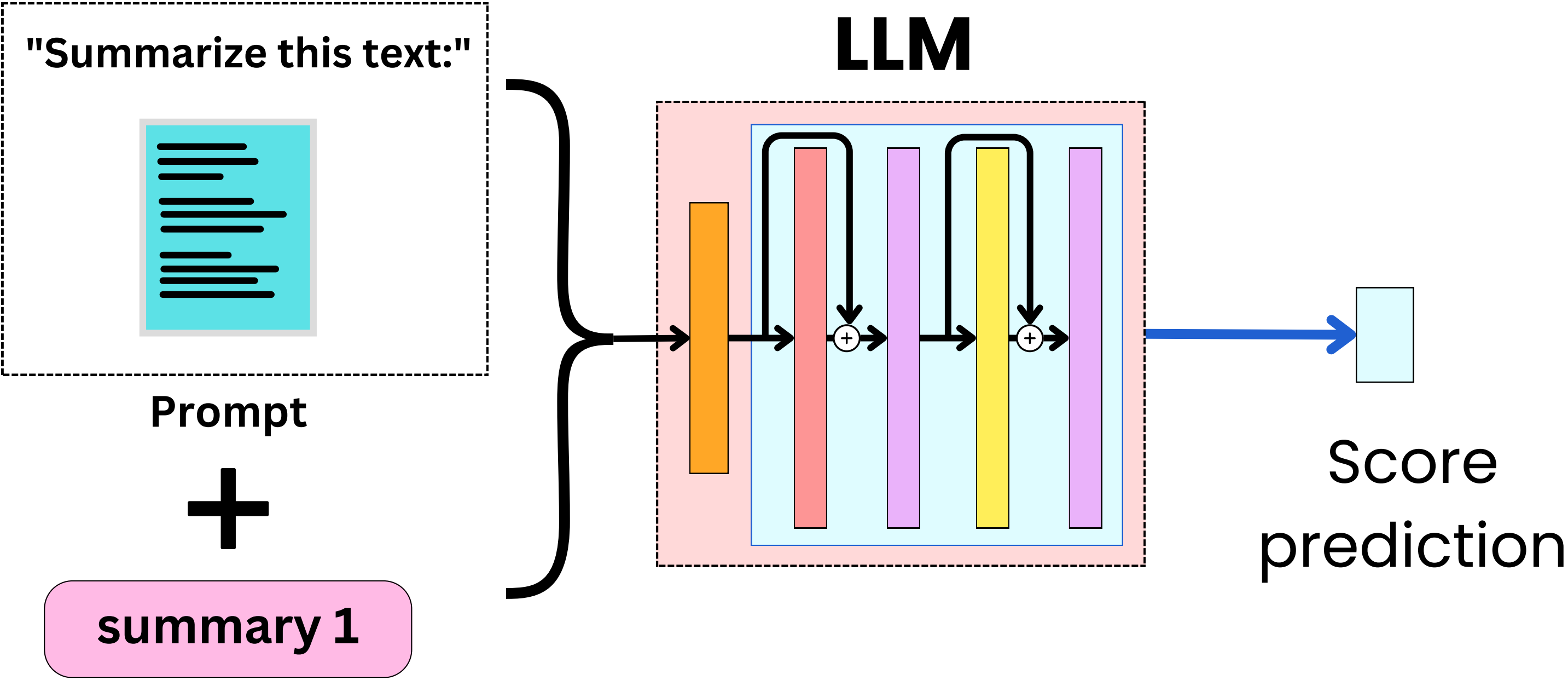
The Reward Model

Rank model outputs with human labeler





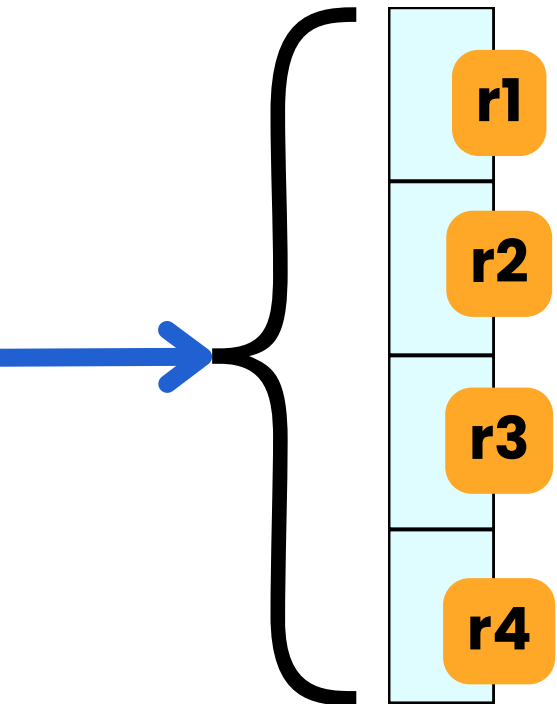
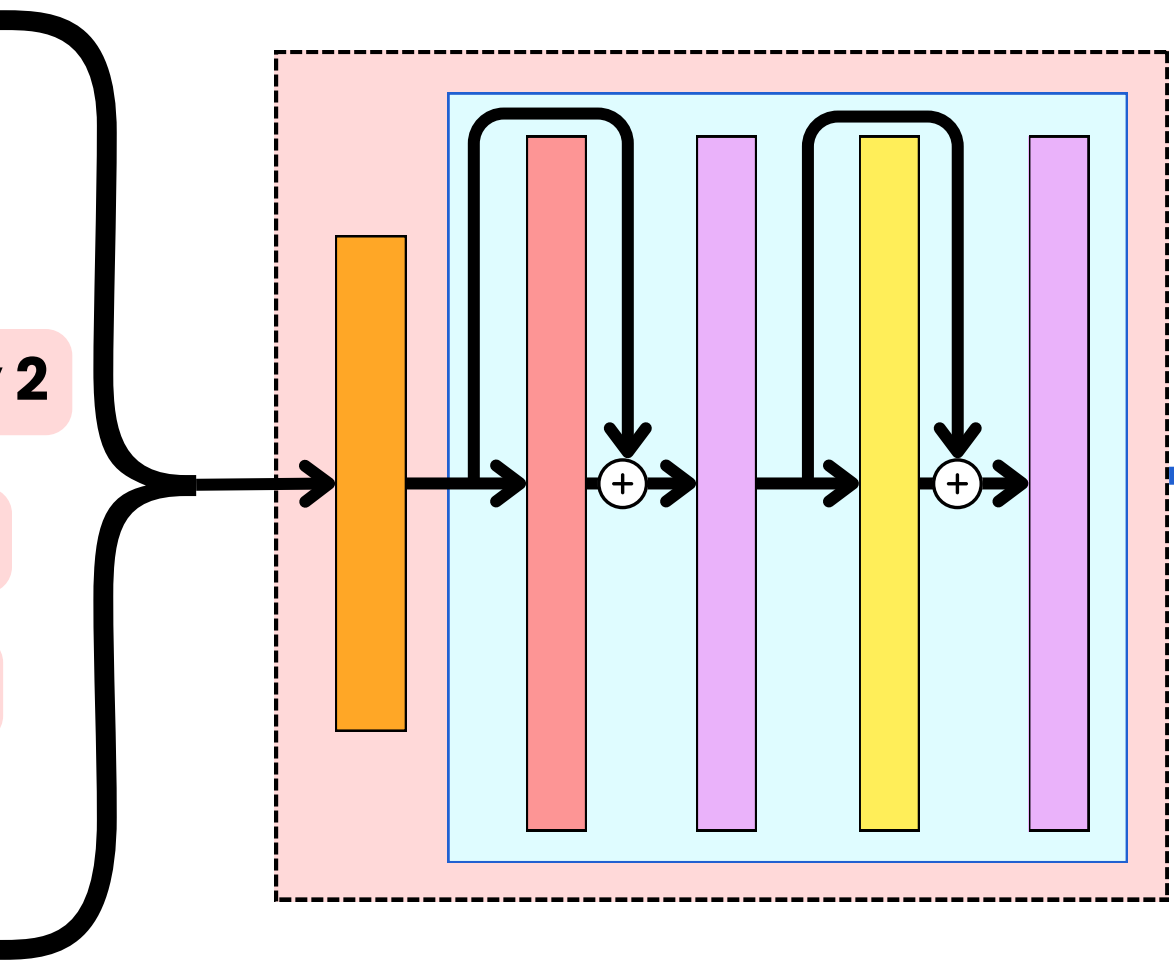
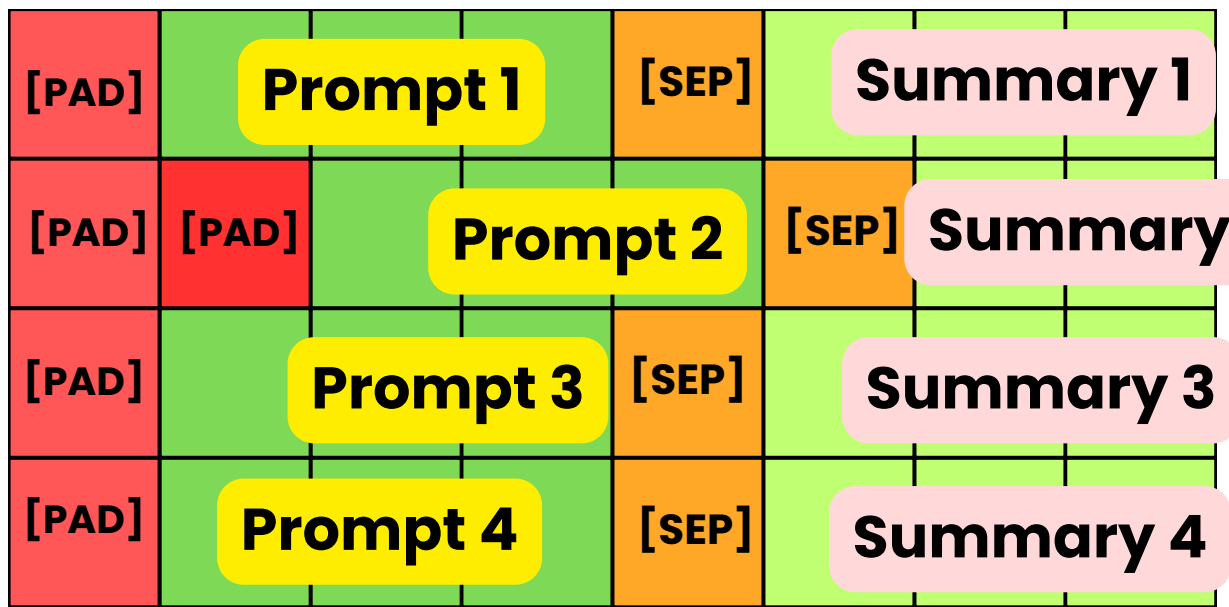
The Reward Model





The Reward Model

LLM

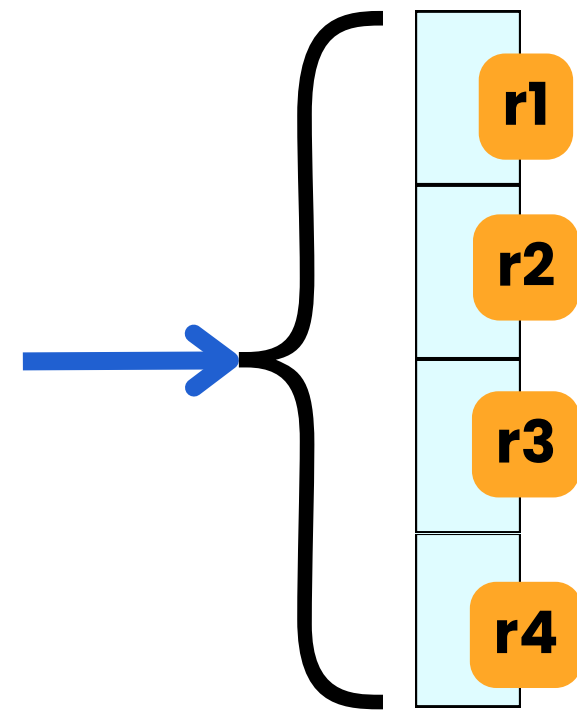


Score
predictions



The Reward Model

summary 1	"4th"
summary 2	"2nd"
summary 3	"1st"
summary 4	"3rd"
Summaries	rank



Score
predictions

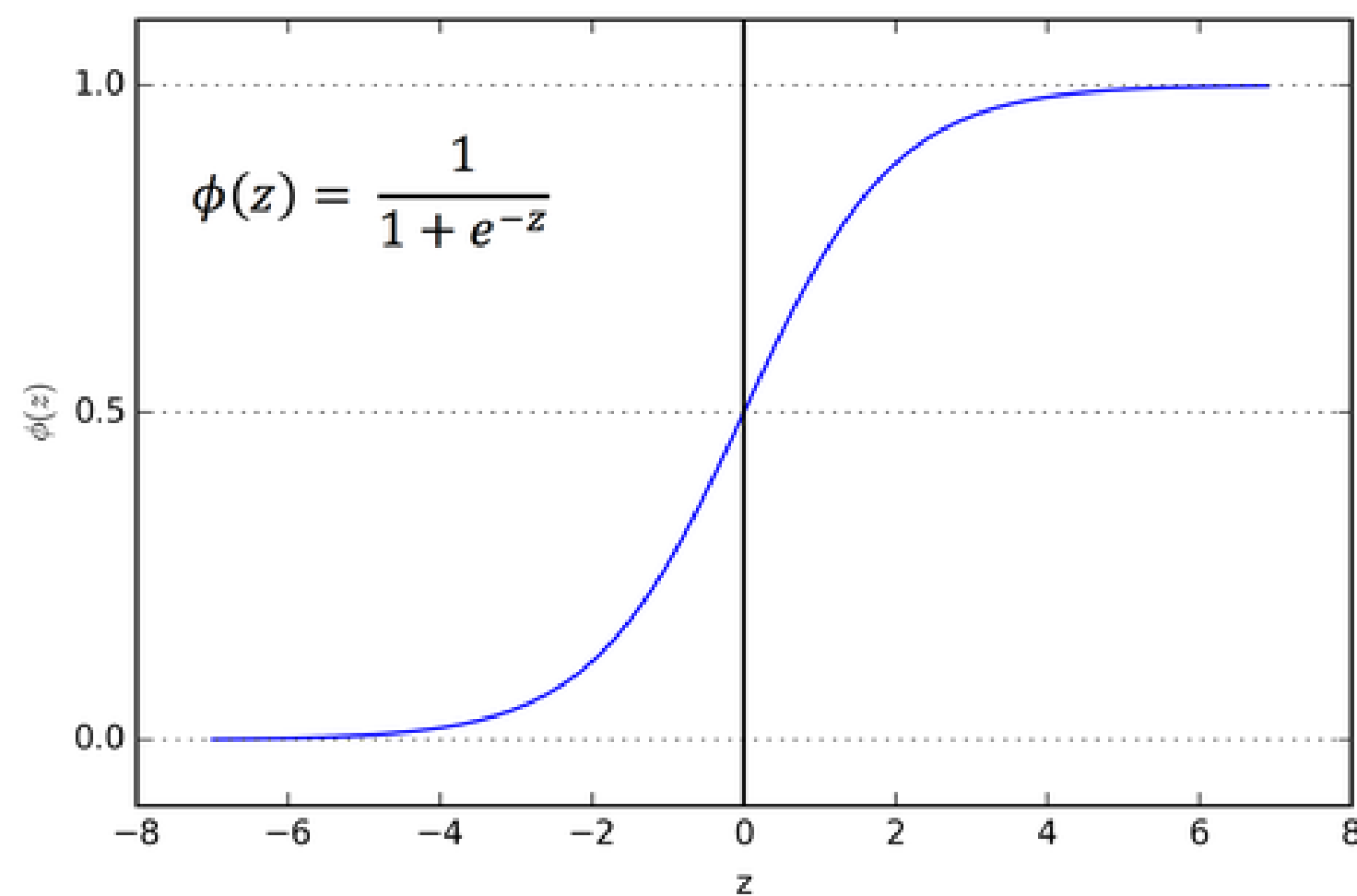
r3	>	r1
r3	>	r2
r3	>	r4
r2	>	r1
r2	>	r4
r4	>	r1



The Reward Model

$$\mathbf{r3} > \mathbf{r1}$$

$$\text{loss}(r_3, r_1) = -\log \left(\frac{1}{1 + \exp -(r_3 - r_1)} \right)$$





The Reward Model

$$\mathbf{r3} > \mathbf{r1}$$

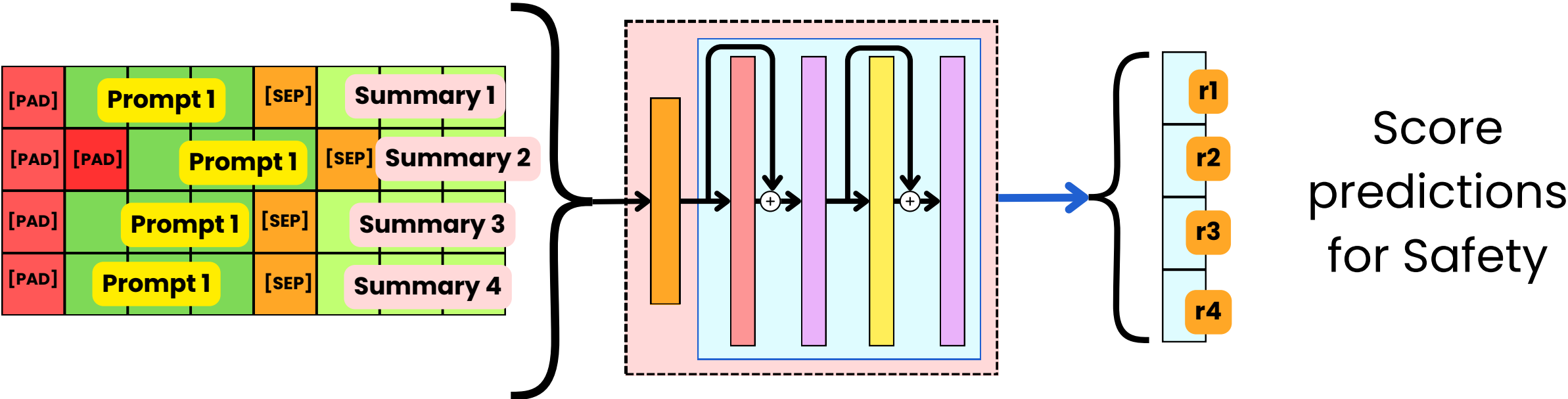
$$\text{loss}(r_3, r_1) = -\log \left(\frac{1}{1 + \exp - (r_3 - r_1)} \right)$$

$$L = \frac{1}{\sum_{r_i > r_j} \mathbb{1}_{r_i > r_j}} \sum_{r_i > r_j} \text{loss}(r_i, r_j)$$

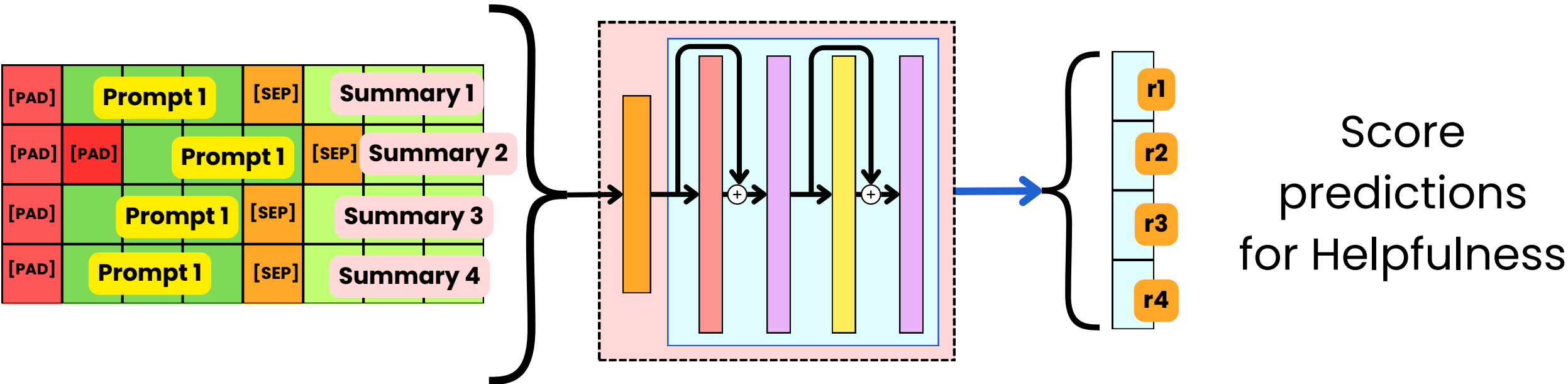


The Reward Model

Reward model 1



Reward model 2





Reinforcement Learning

The intuition

Cross-entropy for
classification

$$-\log P(\text{answer}|\text{prompt})$$

If agrees with label



Reinforcement Learning

The intuition

Cross-entropy for
classification

$$-\log P(\text{answer}|\text{prompt})$$

If agrees with label

Policy gradient
loss

$$-R(\text{prompt} + \text{answer}) \log P(\text{answer}|\text{prompt})$$



Reinforcement Learning

The intuition

Cross-entropy for
classification

$$-\log P(\text{answer}|\text{prompt})$$

If agrees with label

Policy gradient
loss

$$-R(\text{prompt} + \text{answer}) \log P(\text{answer}|\text{prompt})$$

Advantages $A(\text{Prompt} + \text{answer}) = R(\text{Prompt} + \text{answer}) - V(\text{Prompt})$



Reinforcement Learning

The intuition

Cross-entropy for
classification

$$-\log P(\text{answer}|\text{prompt})$$

If agrees with label

Policy gradient
loss

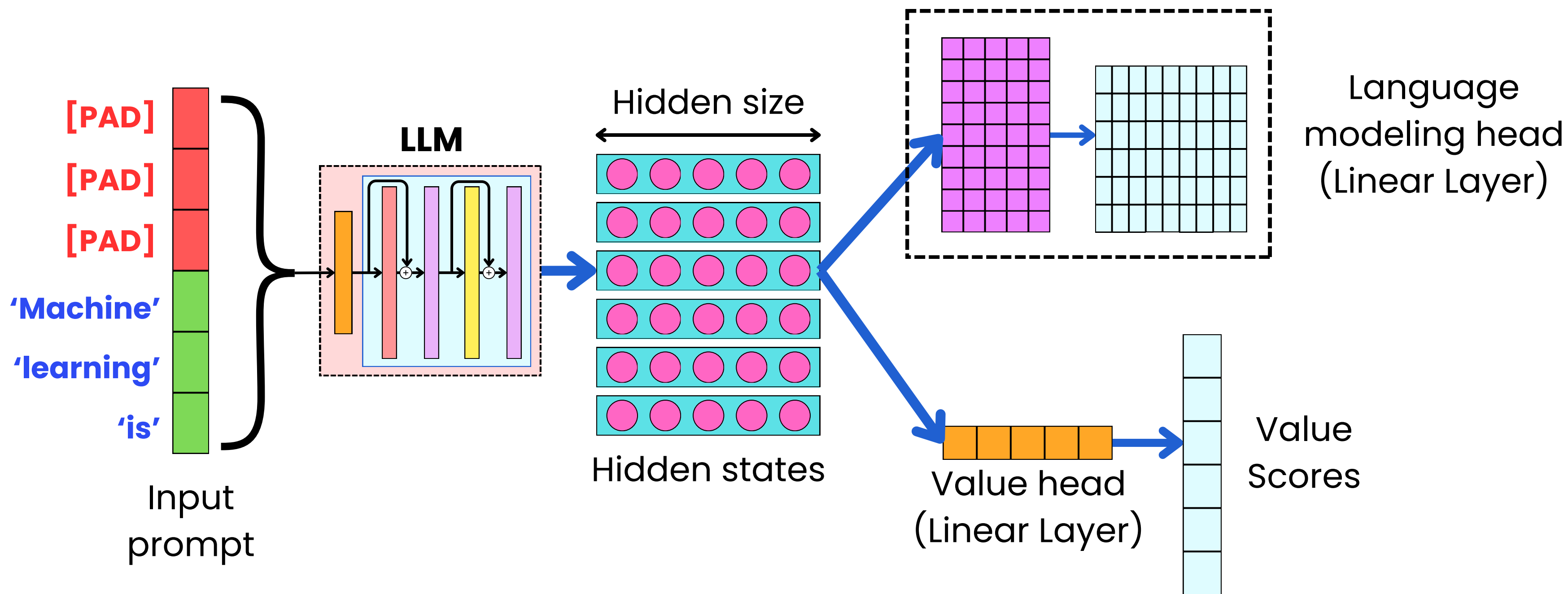
$$-R(\text{prompt} + \text{answer}) \log P(\text{answer}|\text{prompt})$$

Advantages $A(\text{Prompt} + \text{answer}) = R(\text{Prompt} + \text{answer}) - V(\text{Prompt})$

Actor-Critic loss $-A(\text{Prompt} + \text{answer}) \log P(\text{answer}|\text{Prompt})$



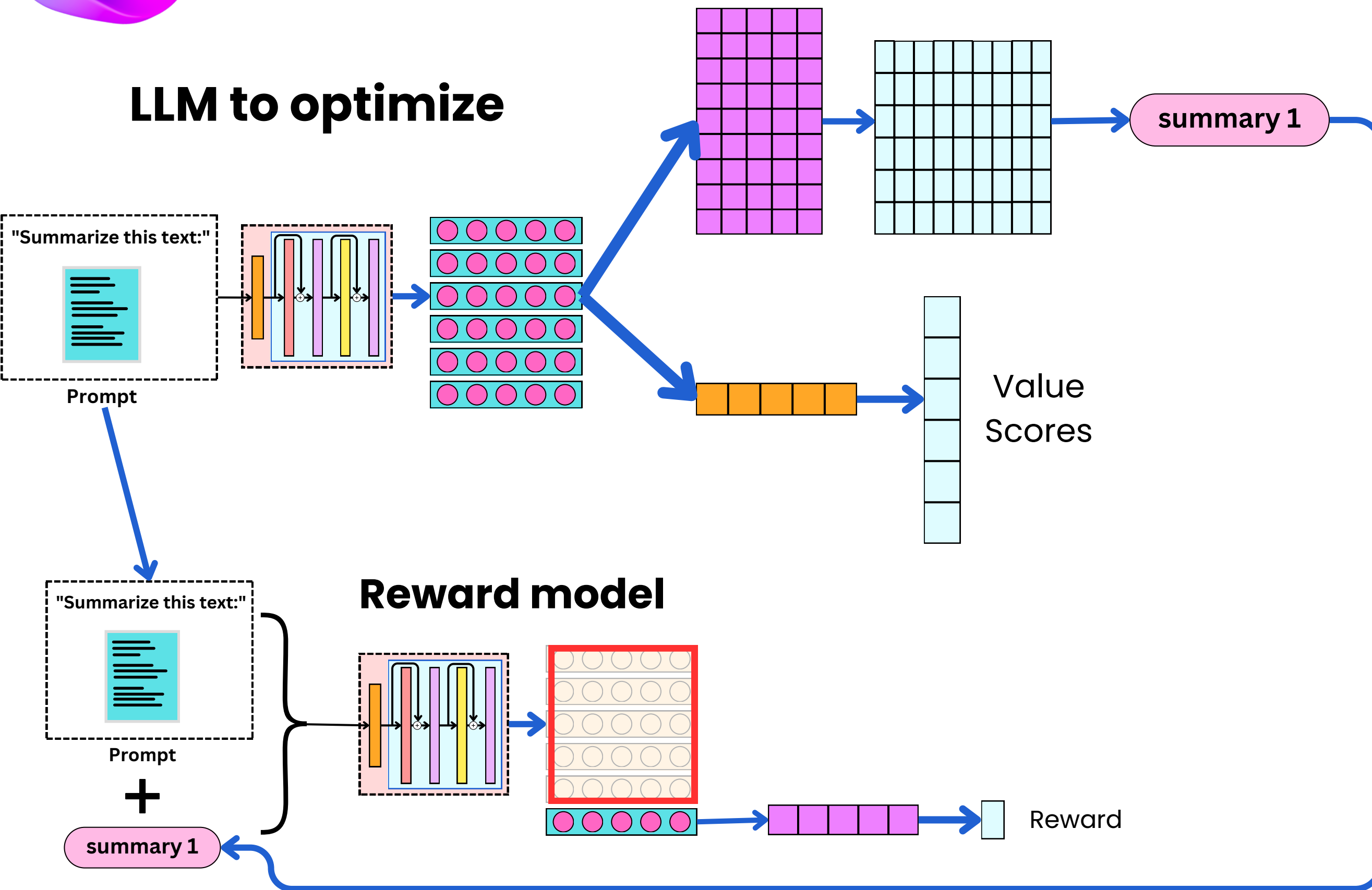
Reinforcement Learning





Reinforcement Learning

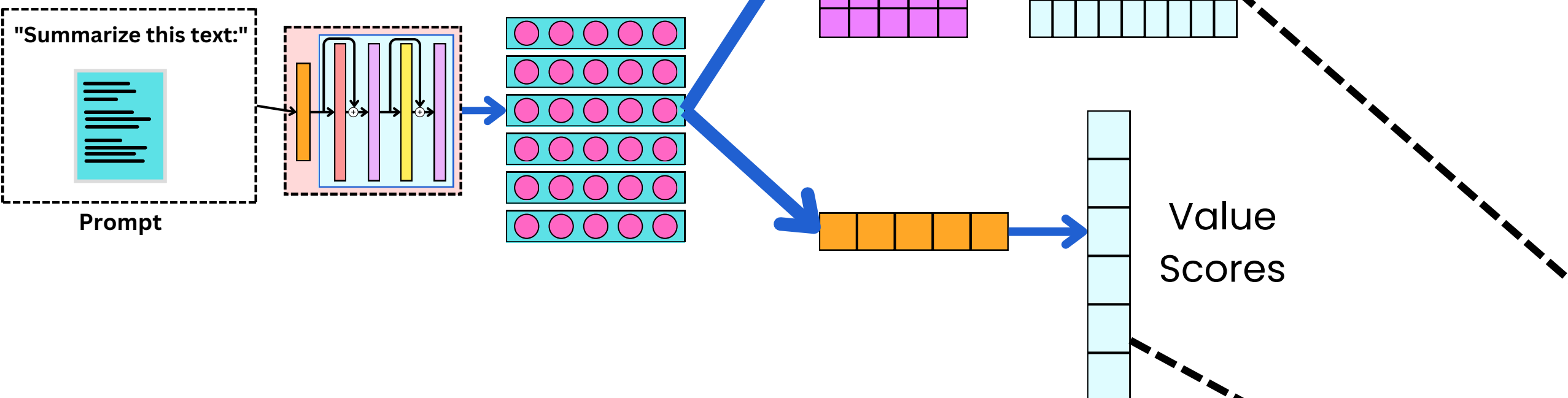
LLM to optimize





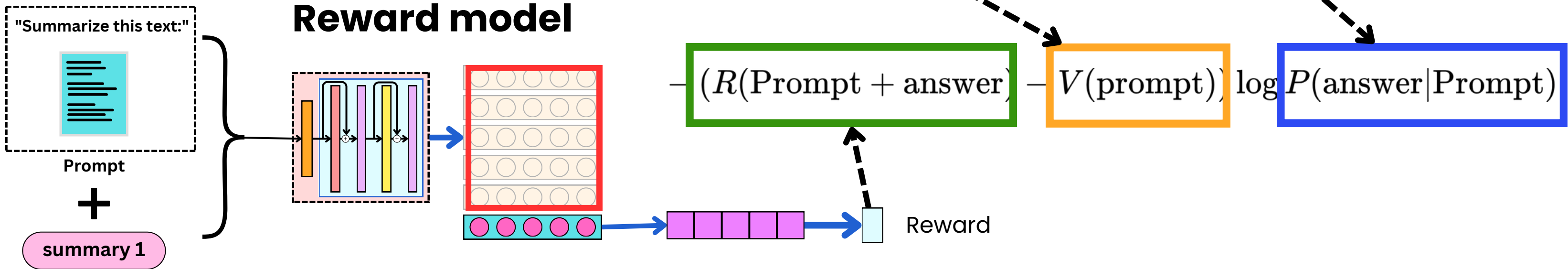
Reinforcement Learning

LLM to optimize



Actor-Critic

Reward model



$$-(R(\text{Prompt} + \text{answer}) - V(\text{prompt})) \log P(\text{answer} | \text{Prompt})$$



Reinforcement Learning

The intuition

Actor-Critic loss $-\left(R(\text{Prompt} + \text{answer}) - V(\text{prompt})\right) \log P(\text{answer}|\text{Prompt})$



Reinforcement Learning

The intuition

Actor-Critic loss $-(R(\text{Prompt} + \text{answer}) - V(\text{prompt})) \log P(\text{answer}|\text{Prompt})$

Trust Region Policy Optimization (TRPO) $-(R(\text{Prompt} + \text{answer}) - V(\text{prompt})) \frac{P_{new}(\text{answer}|\text{Prompt})}{P_{old}(\text{answer}|\text{Prompt})}$

With constraint $\frac{P_{new}(\text{answer}|\text{Prompt})}{P_{old}(\text{answer}|\text{Prompt})} < \delta$



Reinforcement Learning

The intuition

Actor-Critic loss $-(R(\text{Prompt} + \text{answer}) - V(\text{prompt})) \log P(\text{answer}|\text{Prompt})$

Trust Region Policy Optimization (TRPO) $-(R(\text{Prompt} + \text{answer}) - V(\text{prompt})) \frac{P_{new}(\text{answer}|\text{Prompt})}{P_{old}(\text{answer}|\text{Prompt})}$

With constraint $\frac{P_{new}(\text{answer}|\text{Prompt})}{P_{old}(\text{answer}|\text{Prompt})} < \delta$

Proximal Policy Optimization (PPO) $r = \frac{P_{new}(\text{answer}|\text{Prompt})}{P_{old}(\text{answer}|\text{Prompt})}$

$\min(rA, \text{clip}(r)A)$



$$r = \frac{P_{new}(\text{answer}|\text{Prompt})}{P_{old}(\text{answer}|\text{Prompt})}$$

$$L^{CLIP}(\theta) = \hat{\mathbb{E}}_t \left[\min(r_t(\theta) \hat{A}_t, \text{clip}(r_t(\theta), 1 - \epsilon, 1 + \epsilon) \hat{A}_t) \right]$$

